Structured Tactile Representations for Bimanual Contact-Rich Manipulation:

Task Topologies, Physical Grounding, and Evidence

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Abstract

Tactile learning has progressed from task-specific force features to self-supervised encoders, cross-sensor foundation models, spatial anchoring, contact fields, and graph-based dynamics. In parallel, bimanual manipulation has advanced through large-scale simulation, imitation, human-motion retargeting, and increasingly capable real systems. The intersection remains conceptually fragmented: a representation that recognizes local contact is not necessarily one that explains how load, slip, and constraint changes propagate between two hands through an object, tool, workpiece, or environment. This paper presents a curated scoping review of 136 archived records spanning tactile representation, bimanual coordination and handover, relational and physics-informed learning, tool–workpiece control, and datasets. 135 scholarly works enter the narrative synthesis; publication status is kept explicit because the rapidly moving 2025–2026 frontier contains many preprints and programme-listed papers. The archive-wide evidence map is complemented by full-text precision coding of a purposive 30-paper core (19 Level A and 11 Level B records), with located evidence separated from author claims and reviewer inference.

We organize the literature along six independent axes: task/contact topology, representation scale, structural prior, supervision source, deployment observability, and strength of evaluation evidence. This organization exposes three recurring gaps. First, most tactile encoders model local or temporal content without testing whether cross-interface load paths are recoverable. Second, graph and contact-structured methods often obtain their topology from kinematics, simulator state, or predefined edges rather than infer it from deployable observations. Third, bimanual studies commonly report task success without matched-budget representation baselines, transition-level diagnostics, or topology-level out-of-distribution tests. None of the 30 core papers tested strict held-out contact-topology transfer under a deployment-observable inference protocol. Handover provides a controlled load-transfer laboratory, while cooperative tool–workpiece manipulation adds a harder multi-hop topology. Tool use itself is established, but matched studies of observation-grounded cross-hand load flow remain comparatively sparse in the mapped corpus. New full-text audits show that cross-hand tactile relations and compliant force-history control already have direct evidence in different settings. They narrow the unresolved question to matched robotic load-path validation and topology-defined transfer, rather than the existence of relational touch. We conclude with an evidence-based research agenda centred on observation-grounded structure, interface-level prediction, uncertainty, causal interventions, and transfer across contact topologies. The review does not argue that graphs are intrinsically superior; it specifies the experiments required to establish when explicit structure is useful.

Keywords: tactile representation learning; bimanual manipulation; contact-rich robotics; dynamic contact graphs; handover; load transfer; tool use; visuo-tactile learning; scoping review.

Scope statement. This is a reproducible, curated scoping review of the local research archive, not a PRISMA-complete systematic review or a quantitative meta-analysis. Counts describe the archived corpus and should not be interpreted as publication-rate estimates for the entire field.

1 Introduction

Contact-rich manipulation is partially observable in a particularly consequential way. Vision can estimate object pose before contact, but it cannot directly reveal the pressure redistribution that precedes slip, the direction of an incipient constraint, or whether an apparently stable object is actually supported by the left hand, the right hand, a tool, or the environment. Tactile sensing reduces this ambiguity at an individual interface. Bimanual manipulation then creates a second problem: the controller must relate several local interfaces whose states are coupled through shared rigid-body motion, compliance, friction, and changes in contact topology.

The distinction is easiest to see in a handover. Before transfer, the giver bears most of the load. During dual contact, the same object couples both hands, and a safe release depends on the receiver's support rather than on giver touch alone.

After release, the cross-hand path disappears. A representation that concatenates two tactile streams can, in principle, approximate this mapping, but it does not state which interfaces interact, which entity mediates the interaction, or why the mapping should transfer to a new grasp topology. Conversely, a graph can name the interfaces, yet a graph whose edges are fixed or supplied by a simulator does not solve contact inference at deployment.

Recent work makes this review timely. Large tactile pretraining systems learn sensor-robust local features from video-based touch, robot play, and multimodal alignment [20, 23, 28, 100, 115]. Spatially anchored and cross-modal models seek to place touch in a geometric or temporal context [26, 32]. Relational models use graphs for tactile arrays, learned dynamics, morphology, and contact physics [1, 4, 21, 71, 102]. At the same time, bimanual policy learning has expanded from simulated suites to human demonstrations and real systems [9, 22, 36, 43, 74]. The 2026 frontier further includes tactile-reactive bimanual control, whole-hand benchmarks, morphology-aware tactile policies, contact-anchored control, and tool-oriented contact fields [17, 33, 52, 63, 107].

The two late-August surveillance updates sharpen this frontier. Hierarchical tactile world models now separate contact, deformation, and slip prediction; sequential visuo-tactile learning uses temporal alignment and training-time privileged posteriors; and physics-aware sensing explicitly models object properties or time-dependent viscoelastic relaxation [50, 82, 92, 96]. Meanwhile, structured cloth dynamics combine a mesh prior with a learned global residual [84]. These advances strengthen the relevant alternatives to a contact graph: a valid comparison is no longer "graph versus concatenation", but observation-matched structured priors, predictive temporal models, and residual global models. None of these additions, however, directly demonstrates observation-grounded cross-hand load topology.

The September–early-October catch-up changes the wording of the gap itself. BiView-Touch provides direct offline human cross-hand tactile representation evidence, while temporal tactile encoding with compliance provides a strong real bimanual handover control baseline [46, 56]. CADeT additionally shows observation-grounded transmission-mode inference and physical probing in visually observed tissue manipulation [30]. These are positive results at different links of the evidence chain. The remaining question is their combination with independently validated robotic cross-hand loads, matched structural controls, and topology-defined transfer, rather than whether relational touch or inferred interaction modes exist at all.

These advances do not yet answer a shared representation question:

> *When does an explicit, observation-grounded model of contact topology and load propagation improve bimanual perception or control over an equally capable unstructured temporal model?*

This formulation avoids two shortcuts. The first is to equate tactile input with physical grounding. A policy may receive fingertip forces while depending on object state, simulator contacts, or a future reference trajectory. The second is to equate a graph architecture with a scientific contribution. Generic relational learning is well established [4, 81, 105]; the contribution must lie in what the nodes and edges mean, how they can be inferred, which physical quantities they predict, and which counterfactual tests they survive.

1.1 Review objectives and contributions

This review integrates literatures that are often treated separately: local tactile representation learning, structured contact and dynamics models, bimanual manipulation and handover, cooperative tool use, and benchmark design. Its contributions are fourfold.

1. It introduces an orthogonal taxonomy that separates task topology, representation level, structural prior, supervision source, deployment observability, and evaluation evidence. This prevents a method from appearing physically grounded solely because it uses tactile inputs or graph layers.
2. It distinguishes *local contact representation* from *cross-interface coordination representation*. The former encodes what a sensor sees; the latter must preserve which interfaces are coupled, how support is redistributed, and how this relation changes over time.
3. It positions quasi-static handover and cooperative tool–workpiece tasks as complementary experimental topologies. Handover isolates support transfer; tool use tests multi-hop, role-asymmetric coupling without making an unsupported claim that the area is empty.
4. It converts broad calls for "better physical representations" into a falsifiable evaluation programme: matched observation and parameter budgets, oracle, inferred, and random structure comparisons; contact-transition metrics; object-, session-, and contact-sequence-level splits; uncertainty calibration; and targeted interventions.

The intended outcome is not a leaderboard. Sensor form factors, embodiments, training data, task semantics, and success definitions vary too much for a valid pooled effect estimate. Instead, the review identifies which claims can be made from current evidence and which experiments are still needed.

1.2 Terminology

We use *tactile observation* for a measured taxel array, tactile image, deformation field, event stream, or calibrated interface force/torque signal. Simple simulator contact flags and full object state are called *privileged contact variables* unless the deployment system measures them. *Contact topology* denotes the set of active interfaces between hands, objects, tools, workpieces, and the environment. *Load path* denotes a sequence of such interfaces through which forces and moments are transmitted. *Structured representation* covers graphs, contact fields, object-centric frames, spatial anchors, and other models that state relations beyond token order. These terms describe information and assumptions, not a preferred neural architecture.

2 Review Method and Corpus

2.1 Review design

The study is a curated scoping review. Scoping review methodology is appropriate when concepts, evidence types, and evaluation practices are heterogeneous and the goal is to map a field rather than estimate one pooled treatment effect [78]. The accompanying `review_protocol.md` records the scope, eligibility criteria, evidence fields, update procedure, and limitations. The machine-readable `evidence_matrix.csv` is generated from the workspace manifest by a versioned script; rule-assisted annotations remain subject to manual audit. A second evidence layer, stored in `precision_annotations/`, contains full-text precision coding and an individual evidence card for each of 30 purposively selected core papers.

The initial corpus snapshot was frozen on 12 August 2026. It began from a decision-oriented archive assembled for research planning and was expanded around five intersecting areas: (i) tactile and visuo-tactile representation learning; (ii) bimanual dexterous manipulation and handover; (iii) graph, contact-field, object-centric, and physics-informed models; (iv) tool–workpiece contact control; and (v) datasets and benchmarks. Backward and forward chaining was used selectively for landmark and directly competing works. Author project pages, arXiv records, official conference programmes, proceedings pages, and publisher records were used to disambiguate status.

Two rapid-surveillance passes covering 20–26 August and 27–30 August 2026 identified 17 additional eligible studies. These records were screened into the archive-wide layer, their public full texts were retrieved where available, and their implications were integrated into the narrative synthesis. The manuscript corpus was therefore updated and refrozen on 30 August 2026 at 91 records. The 30-paper precision subset was not silently enlarged: the new records have the lighter archive-wide coding described below, and all claims based on the precision subset retain a denominator of 30.

A catch-up pass on 7 October 2026 added 14 records after located full-text checks: 13 first released after 30 August and one older work, ContactWorld, newly discovered through a September revision. The new studies' inputs, structural assumptions, evaluation splits, metrics, and limitations were audited in primary full texts and integrated into this narrative. The manuscript was refrozen at 105 records on 7 October; precision coding remains dated 13 August. The dated screening note records versions, search bounds, watchlist leads, exclusions, and evidence locations. This was a targeted catch-up, not a claim of complete coverage through the end of 7 October.

A further surveillance pass on 14 October adds 16 located full-text archive audits after screening releases/revisions after 7 October. TouchScale is explicitly a 7 October first release with an 8 October revision, not a new first release in the interval. After narrative integration, the manuscript is refrozen at 121 records. Precision coding remains 13 August.

A 21 October pass adds 15 further full-text-audited studies, updates ForeTac's v2 method description, and verifies ILA's presentation-only v2 change. After narrative integration the corpus is refrozen at 136 records. The dated ledger separates post-14-October releases from boundary/backlog leads and selected existing-paper revisions. The precision date and denominator remain unchanged.

The current draft follows the reporting intent of PRISMA-ScR but does not claim formal PRISMA compliance. It does not reproduce database-by-database search strings, use dual independent screening, or apply a validated cross-design risk-of-bias tool. These omissions matter: the numerical corpus summary is an audit of this archive, not a bibliometric description of the whole field.

2.2 Two-layer evidence design

The archive-wide layer covers all 136 records. Title, abstract, publication and artifact metadata, and selectively consulted full text support eligibility, status accounting, and coarse six-axis coding. This layer provides breadth but is not used as if every row were an independently extracted full-text study. The 45 records admitted across the October updates have a separate located full-text archive audit that overrides coarse coding; this intermediate audit does not silently promote a paper to the precision subset or constitute independent reproduction.

The precision layer was completed on 13 August 2026 for 30 papers selected for their decision relevance rather than for statistical representativeness. The subset comprises 19 Level A records, coded through full text plus available artifact

Table 1: Review questions and the evidence required to answer them.

RQ	Question	Primary evidence fields
RQ1	What must be preserved for bimanual contact-rich manipulation?	Task topology, predicted quantities, temporal horizon, interface identity.
RQ2	How is touch organized across sensors, bodies, entities, and time?	Representation scale, tokenization, spatial frame, structural prior.
RQ3	Is the structure available at deployment?	Sensor inputs, privileged state, inferred versus supplied topology, future reference.
RQ4	How is usefulness demonstrated?	Baselines, split unit, probes, control metrics, OOD tests, interventions, uncertainty.
RQ5	Where are the actionable gaps?	Missing topology–method cells, weak evidence links, unavailable labels or benchmarks.

or code evidence, and 11 Level B records, coded through full-text method and evidence analysis. It covers direct system competitors, executable benchmarks, graph and relational models, contact fields and anchors, mechanics and control references, tool-use systems, and candidate local tactile encoders. Each record distinguishes (i) the authors' claim, (ii) evidence located in the paper or artifact, and (iii) reviewer inference. It also separates deployment inputs from training-only privilege and codes object, sensor, embodiment, and contact-topology OOD as different claims. All 30 cards contain evidence locations; the procedure is full-text evidence coding, not independent reproduction.

2.3 Eligibility and coding

Scholarly work was included when it materially addressed robot touch, contact-state or dynamics estimation, tactile-informed control, bimanual coordination, handover, tool use, or an enabling benchmark. Generic graph and physics-learning papers were included only where their formulation directly informs relational contact modelling. A community tactile-data index was retained as a resource record but excluded from study-level synthesis. Duplicate versions were represented by the most informative accessible record, with publication status documented rather than silently upgraded.

Each record was coded along six axes:

1. **Task/contact topology:** single interface, multi-finger hand, two hands sharing an object, sequential/overlapping handover, hand–tool– environment, or hand–tool–workpiece–hand.
2. **Representation scale:** local patch, finger/link, hand, object/interface, cross-entity relation, or trajectory-level latent.
3. **Structural prior:** token order, kinematics, object-centric coordinates, spatial anchors, contact fields/modes, or graph connectivity.
4. **Supervision source:** instance discrimination, reconstruction, cross-modal alignment, action-conditioned prediction, simulator labels, force/pose measurement, or task reward.
5. **Deployment observability:** directly sensed, perception-derived, training-only privilege, or unavailable future reference.
6. **Evidence:** representation probe, open-loop prediction, closed-loop control, real transfer, OOD topology, intervention, and uncertainty.

The axes are kept separate because they answer different validity questions. A real-robot paper can use weak splits; a simulation paper can isolate a causal mechanism cleanly; a high-capacity pretrained encoder can be deployable but offer no cross-interface structure; and a physically interpretable graph can depend on unavailable state.

2.4 Corpus accounting and status discipline

The archive contains 136 records: 15 core or directly competing systems, 30 bimanual and handover works, 44 tactile-representation works, 25 graph, physics, and contact works, 11 tool–workpiece works, and 11 dataset or benchmark records. 135 scholarly studies enter the narrative synthesis and one community index is resource-only.

The recorded August cache comprised 86 validated PDFs and five source-only entries. The 7 October pass verified three selected PDFs. The 14 October admissions use fourteen official HTML full texts and two official PDFs; the entire cache was not refreshed. The 21 October additions use fourteen official HTML full texts and one official PDF; ForeTac and ILA exact-version PDFs were compared separately. The 131 current PDF URLs and five source-only entries describe retrieval routes, not the number of locally present or freshly validated files.

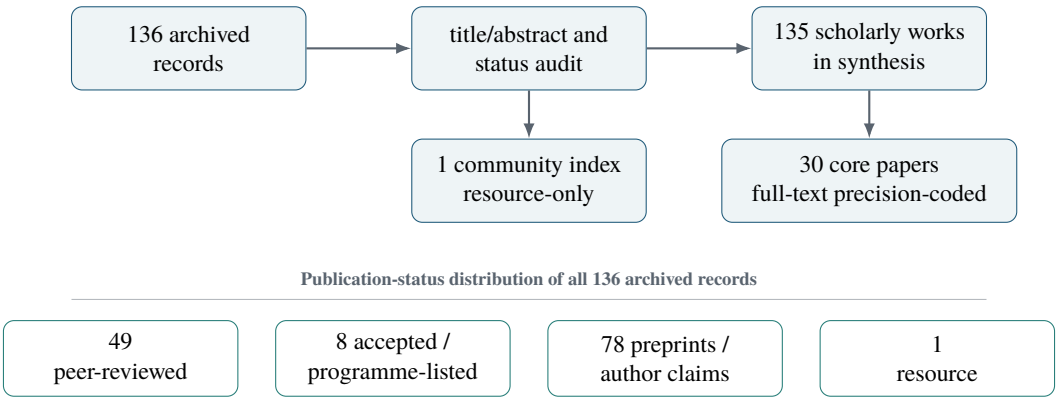


Figure 1: Two-layer corpus flow and publication-status accounting. All 136 records receive archive-wide screening; 45 additions also have located full-text archive audits. The purposive 30-paper subset receives the separate precision coding. Counts describe the curated archive, not the global literature.

The regenerated evidence matrix records 49 peer-reviewed works, eight works listed as accepted or on an official programme, 78 preprints or author-claim records, and one community resource. All 45 records admitted across the October updates remain in the last scholarly category: author-reported acceptance, workshop presentation, submission, and a journal-style running header do not establish a final publication record. The inherited TactiDex author-page acceptance label is now assigned to the author-claim tier; its raw status is preserved. Other historical status labels are not a claim of a new venue-by-venue audit.

2.5 Synthesis strategy

The synthesis is concept-centric rather than paper-by-paper. Representative methods are compared within the same information problem, and negative space is made explicit: for example, a work may demonstrate cross-sensor transfer without testing cross-hand load inference, or closed-loop bimanual success without an explicit tactile representation ablation. No pooled accuracy or success rate is reported because task definitions, sensor density, action spaces, data scale, embodiments, and test splits are not commensurate. Descriptive counts from the precision subset are reported only when the denominator and coding rule are explicit. In particular, zero of the 30 core papers evaluated strict held-out contact-topology OOD under deployment-observable inference. Here "strict" means that the participant/edge structure itself is held out; unseen objects, contact modes within a fixed structure, tool geometry, or semantic task compositions do not qualify. This 0/30 finding is not recomputed for the expanded archive. Claims that a specific intersection is comparatively sparse are stated as inferences from the mapped corpus rather than field-wide publication counts.

3 Task Topologies and Representation Requirements

3.1 Why topology is a more useful organizing variable than task name

Task labels conceal important physical differences. "Handover" can mean a quasi-static transfer with simultaneous support, a dynamic throw–catch event, or a robot–human release triggered by pull and slip. "Tool use" can mean one hand moving a rigid tool against a fixed environment, one hand stabilizing a workpiece while the other acts, or both hands jointly holding the same tool. These variants demand different information even when their semantic label is the same. A representation-oriented review should therefore begin with the active interfaces and their changes.

Let the task at time t contain entities $\mathcal{V}_t = \{H_L, H_R, O_1, \ldots, T_1, \ldots, W, E\}$ for left and right hands, objects, tools, a workpiece, and environment. The physical contact topology is $\mathcal{G}_t^c = (\mathcal{V}_t, \mathcal{E}_t^c)$, where an edge represents an active load-bearing or constraint-producing interface. This notation does not require the model itself to be a graph neural network. It provides a neutral way to ask whether the observation history $o_{1:t}$ contains enough information to infer the relevant interfaces and their direction-dependent effects.

3.2 Six topology families

Local single-interface contact. The simplest case associates a tactile patch with local geometry, force, slip, or material. This is the dominant pretraining unit for image-based tactile encoders. It supports transferable perception but leaves global interface identity and cross-body coupling to downstream models.

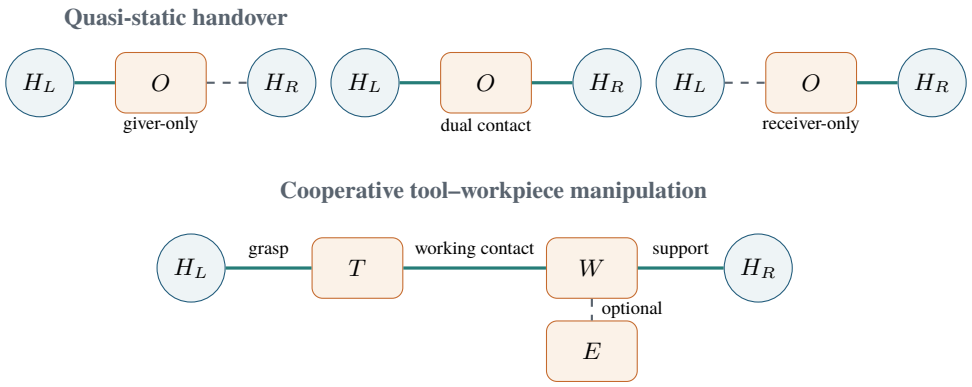


Figure 2: Two complementary topology families. Handover changes which hand supports a shared object; tool–workpiece manipulation introduces a longer, role-asymmetric load path. Dashed edges are inactive or optional.

Multi-finger contact within one hand. In-hand manipulation requires several contacts to be fused across finger kinematics and time. Distributed-taxel graphs and hierarchical tactile graphs explicitly exploit this structure [21, 102]. The object couples the fingers, making this topology a useful precursor to the two-hand problem, but it does not include inter-hand role changes or object transfer.

Two hands sharing one object. Both hands may grasp, reorient, stretch, or stabilize the same object. Bimanual simulation suites and real policy-learning systems cover many such tasks [9, 36, 74]. Success can arise from kinematic coordination and demonstration priors even when load sharing is not explicitly represented.

Handover. Handover is a topology transition rather than merely a motion trajectory. Static and quasi-static transfer offers giver-only, dual-contact, and receiver-only segments; dynamic throw–catch adds a flight phase and impact. The latter is an impressive control test [31] but a weaker primary benchmark for continuous cross-hand load propagation because the hands are not simultaneously connected during flight. Controlled overlapping handover is better suited to studying support redistribution, release readiness, and transition uncertainty [15, 85].

One hand, tool, and environment. Tool contact creates extrinsic interfaces whose geometry and compliance may be distant from the tactile sensor. Dynamics models, contact-mode controllers, and contact fields address this setting [27, 65, 75, 79]. It tests whether local hand sensing can identify remote tool contact, but not necessarily whether two hands coordinate.

Two hands with a tool and workpiece. Here one hand actuates a tool and the other stabilizes or repositions the workpiece. Brushing a plate, wiping an object, scrubbing, cutting, peeling, or inserting against a held fixture produces a multi-hop path such as $H_L \to T \to W \to H_R$. BUDS establishes the importance of stabilizer/actor role assignment [22]; TACO documents broad bimanual tool–action–object semantics [51]; and TAMEn includes tactile-aware data collection in tasks such as dish washing [89]. Tool use is therefore an established area rather than an uncrowded task category. The narrower mapped cell—deployment-observable cross-hand load propagation with a held workpiece and matched representation alternatives—contains comparatively few direct studies.

3.3 What a useful representation must retain

Across topologies, five information requirements recur.

1. **Interface identity and quality.** A representation must preserve which finger/link touches which entity, contact confidence, sensor quality, saturation, and missing-data masks. Treating a damaged taxel as zero pressure confounds absence of contact with absence of measurement.
2. **Common spatial meaning.** Local tactile features require a link, hand, object, or task frame before contacts on separate bodies can be compared. Spatial anchoring [32] and object-centric contact fields [27, 52] are different solutions to this frame-assignment problem.
3. **Temporal event structure.** Contact onset, load acceptance, incipient slip, release, impact, and recontact occur on different time scales. A pooled window may recognize a state while obscuring its transition.

Table 2: Topology-specific representation questions. "Direct" means that an interface is measured locally; "remote" denotes an interface whose effects must be inferred through another entity.

Topology	Critical transition	Observable evidence	Core representation test
Multi-finger in-hand	stick–slip / regrasp	several direct finger contacts	Can local evidence be aligned through kinematics and object motion?
Shared-object bimanual	unilateral–bilateral support	two hand streams, proprioception, optional vision	Is cross-hand dependence captured beyond concatenation?
Quasi-static handover	giver–dual–receiver	load redistribution and contact onsets	Can phase, load share, and safe release be predicted without privileged contacts?
Dynamic handover	release–flight–impact	pre-release contact and post-impact impulse	Does the representation anticipate impact and recovery under temporal discontinuity?
Single-hand tool use	free–working contact	grasp touch plus remote tool response	Can extrinsic tool contact be localized and its mode inferred?
Cooperative tool use	support/actor coupling changes	direct hand contacts; remote tool–workpiece effects	Does structure transfer across a multi-hop, role-asymmetric topology?

4. **Cross-interface dependence.** The representation should allow a downstream model to distinguish coincident local events from a mechanically mediated relation. For handover, receiver loading should predict giver unloading and release readiness; for tool use, tool–workpiece changes should affect the stabilizing hand.

5. **Uncertainty and observability.** Contact edges and remote loads are latent. A credible model should express uncertainty when sensors are occluded, saturated, delayed, or out of distribution instead of returning an overconfident topology.

This topology view also clarifies what is *not* required. A model need not reconstruct every Newton of the complete scene wrench to support safe control; relative load share, future interface wrench, slip risk, or a calibrated contact phase may be sufficient. Conversely, high task success alone does not show that the learned latent encodes any of these quantities. Representation claims need diagnostic evidence.

4 Tactile Observations, Sensors, and Data Regimes

4.1 The sensor is part of the representation problem

Tactile learning spans RGB or marker-based deformation images, elastomer flow, taxel pressure arrays, event-based tactile cameras, distributed fingertip force, whole-hand skins, and wrist or interface wrench. These signals differ not only in dimension but in invariances, latency, hysteresis, saturation, calibration, and spatial support. A "tactile token" is therefore not a sensor-neutral primitive. Cross-sensor learning must decide which information is shared—contact geometry, normal/shear deformation, dynamics, semantic material cues—and which remains sensor- or embodiment-specific.

Hardware lifetime is part of this invariance problem. A replaceable-cartridge vision-based fingertip reports markedly improved resistance to abrasion and concentrated loading under its accelerated tests [16], while a uniformly illuminated visuo-tactile sensor couples depth reconstruction with spatiotemporal contact-state and slip-direction classification [54]. Durability and illumination stability do not by themselves solve representation transfer, but degradation state, cartridge identity, and lighting/calibration changes should be recorded as sensor-domain variables rather than absorbed into unexplained model error.

Vision-based tactile sensing has enabled large-scale self-supervision because a frame can be processed with mature visual backbones. Sparsh learns general touch representations from large unlabeled tactile collections [28]; T-Dex uses robotic play to pretrain features useful for dexterous control [23]; UniTouch and AnyTouch align heterogeneous sensors and static/dynamic content [20, 100]; and T3 studies transferable tactile Transformers [115]. These works make local tactile content more reusable, but most cross-sensor objectives do not preserve a global hand–object–hand contact identity by themselves.

Distributed and whole-hand sensors shift the difficulty. A fingertip array has a natural patch boundary; a skin spanning

phalanges and palm requires morphology and sensor-layout metadata. HTT explicitly addresses heterogeneous tactile sensors [5]; HT-Bench expands full-hand tactile representation with egocentric vision [33]; FTP-1 introduces morphology-aware tactile tokens across sensors and embodiments [107]. Such work is directly relevant to bimanual transfer because the two hands may have different coverage, calibration, or missing channels.

4.2 Force estimates, measured wrench, and ground truth

Three quantities are frequently conflated:

- a raw tactile deformation or taxel field;
- a wrench estimated from that tactile signal by calibration or a learned model; and
- an independently measured interface wrench from a dedicated force/torque sensor or instrumented object.

The distinction determines what can serve as a target. Training a model to predict an estimate derived from the same tactile image measures compression or consistency, not independent physical validity. Likewise, a simulator's net fingertip contact force is useful supervision but is not equivalent to a real tactile array. PhysGraph, for example, studies graph-structured bimanual policy learning with simulator state and fingertip resultant forces rather than raw tactile images [44]. This makes it relevant to relational control while limiting conclusions about tactile representation transfer.

Force inference can also drift even when the contact geometry appears unchanged. Relaxation-aware soft-gripper sensing uses onboard vision, infrared thermography, and a temperature-coupled viscoelastic model to estimate force during sustained holds [82]. It shows why contact duration, temperature, and loading history belong in the observation audit. An instantaneous tactile or force frame may be insufficient for a long hold, and agreement with a same-frame proxy should not be mistaken for relaxation-robust force estimation.

For load-transfer research, the strongest target hierarchy is:

1. independent instrumented-object or interface force/torque where feasible;
2. calibrated robot wrist/hand wrench with explicit coordinate transforms and gravity compensation;
3. simulator wrench for pretraining or privileged-to-observation distillation, followed by a real validation subset;
4. proxy load share or contact phase, with the proxy's construction and circularity documented.

4.3 Data regimes and their affordances

Human-collected datasets can provide semantic and visual diversity at scale. Touch and Go aligns human-collected touch and vision [99]; OAKINK2 captures complex bimanual hands–object activity [109]; and TACO focuses bimanual tool–action–object understanding [51]. Their strength is breadth, while precise synchronized robot proprioception, calibrated loads, and intervention labels may be absent.

Robot-collected data provides action, embodiment, and timing. VTDexManip offers a visual-tactile pretraining and dexterous manipulation benchmark that includes bimanual handover [49]. T-Rex reports a large tactile-reactive robot dataset and bimanual capabilities [63]. RCT targets robot-collected touch–vision–language generalization [24], while TactiDex and HT-Bench expand real, whole-hand tactile evaluation [33, 60]. As of the artifact audit on 8 August 2026, the TactiDex author project page stated ACM MM 2026 acceptance, but no downloadable dataset, code, or model artifact had been confirmed; it should not be assumed to be an executable first-month substrate. Because several of these appeared as preprints or accepted works near the corpus freeze, exact public availability, license, and final metadata must be checked before they are treated as reproduction substrates.

Two new datasets illustrate complementary physical labels. ViTacPhys uses human visual–tactile demonstrations to estimate mass class, friction class, and stiffness before conditioning an adaptive grasping policy [50]. SoftVTBench synchronizes policy-visible visual–tactile signals with evaluator-only finite-element state and defines a deformation-aware success criterion [37]. The former makes physical-property probes concrete; the latter shows that task completion can co-exist with excessive deformation. Artifact status remains part of the evidence: at the 30 August audit, ViTacPhys code/data were still listed as forthcoming, and the public SoftVTBench release described a smaller dataset version than the current manuscript.

Bench2Dex adds a useful simulation substrate: 26 task–embodiment settings, 12 hand embodiments, approximately 1,300 demonstrations, and repeatable perturbations with stage/terminal-success diagnostics [104]. Its tactile images are mesh-penetration-depth signals at predefined fingertips, not a calibrated physical sensor model. Crucially, the reported policies use RGB and proprioception and are trained per task–embodiment setting. Supported tactile channels and diverse hands therefore establish experimental opportunities, not measured tactile gains or held-out-embodiment transfer. Public code, data, and checkpoint endpoints were verified; full dataset loading and execution were not. The reported perturbations retain the task and object set, and kinematic safety proxies should not be substituted for measured interaction forces.

Table 3: Representative data regimes and what they can and cannot establish for cross-interface representation. Status reflects the corpus freeze.

Resource / line	Regime	Main strength	Caution for bimanual load-path claims
Touch and Go [99]	human-collected vision–touch	broad multimodal diversity	limited robot action, proprioception, and calibrated cross-hand load labels
T-Dex / Sparsh [23, 28]	robot touch / large unlabeled tactile data	reusable local encoders	downstream topology not specified by pretraining objective
VTDexManip [49]	simulated visual–tactile benchmark	reproducible pretraining and handover task	sim observation and contact fidelity must match intended deployment
T-Rex [63]	large real robot tactile data	temporally rich bimanual control evidence	dataset subset, synchronization, and exact labels require audit
OAKINK2 / TACO [51, 109]	bimanual human/object datasets	topology and semantic task breadth	tactile load evidence is not their primary signal
HT-Bench / TactiDex [33, 60]	whole-hand real tactile benchmarks	coverage beyond fingertips	recent status and resource release should be verified per artifact
ViTacPhys [50]	human visual–tactile demonstrations and robot transfer	explicit mass, friction, and stiffness probes	semantic priors and unreleased artifacts must be separated from tactile evidence
SoftVTBench [37]	deformable-object visual–tactile benchmark	independent deformation state and safety-aware success	public release version must be matched to the reported manuscript scale
Tactile Genesis [14]	scalable tactile simulation	layout variation and dense labels	sim-to-real contact appearance and dynamics gap

Simulation is essential for contact labels and topology interventions. Tactile Genesis targets scalable tactile simulation across layouts [14]; generic graph simulators demonstrate how object/particle relations can support long-horizon physical prediction [2, 71]. Simulation can generate edge, force, slip, and counterfactual labels that are prohibitively expensive in the real world. Its risk is an unobserved shortcut: a model may use precise object state, collision geometry, contact impulses, or future reference that will not exist at deployment.

4.4 Synchronization is not a preprocessing detail

Bimanual tactile data is intrinsically multi-rate. Cameras, tactile arrays, joint encoders, wrist wrenches, and policy actions may run at different rates and have different transport delays. A 50–100 ms misalignment can change an apparent causal relation at contact onset or release. Dataset construction should therefore retain raw timestamps, clock source, interpolation policy, sensor-quality masks, and commanded versus measured action times. Synchronization should be evaluated with event-based sanity checks—for example, whether tactile onset precedes an object acceleration that it could physically cause—rather than only by visually aligned plots.

The split unit is equally important. Frame-level random splits leak object, episode, grasp, and contact-sequence identity into the test set. Representation claims require episode-level splits at minimum, with object, session, operator, sensor, embodiment, and contact-sequence holds-outs selected according to the claim. For cross-topology transfer, the complete topology family—not just a motion instance—must be absent from training.

4.5 A minimum interoperable schema

A useful cross-dataset schema should keep (i) raw tactile observations and quality masks; (ii) calibrated per-sensor metadata; (iii) joint states and link poses; (iv) object/tool/workpiece identifiers and optional perception-derived poses; (v) actions and timestamps; (vi) phase, interface, slip, and wrench labels with provenance; (vii) temperature, contact duration, maintenance/calibration state, and material metadata when relaxation or wear matters; and (viii) split membership. Coordinate frames must be explicit. Without this information, a study can compare end-to-end policies, but it cannot reliably compare structured interface representations.



Figure 3: Representation hierarchy. The arrows denote additional organization, not a mandatory neural architecture. Cross-interface claims require evidence at the rightmost levels, even when the local encoder is strong.

5 From Local Touch to Cross-Interface Representations

5.1 A hierarchy of representation scales

Tactile models can be placed on a hierarchy from sensor-local appearance to task-level physical relations (figure 3). Progress at a lower level is enabling but does not imply progress at every higher level. For example, a sensor-invariant embedding may improve local contact classification while discarding absolute orientation or force magnitude that a load-transfer model needs. Conversely, a task-specific relational policy may model useful interactions while inheriting a fragile local encoder.

5.2 Local self-supervised and transferable touch

Self-supervised tactile learning replaces expensive per-task labels with objectives such as reconstruction, masked prediction, temporal consistency, and instance discrimination. T-Dex couples pretraining to robotic play, showing that unlabeled interaction can benefit downstream dexterity [23]. Sparsh scales self-supervision over vision-based tactile data and evaluates a family of downstream tasks [28]. These approaches establish that generic visual backbones are not enough and that tactile-specific pretraining can be reused.

Transfer across sensors is harder because pixels encode different elastomers, illumination, marker patterns, optics, and spatial scales. UniTouch learns unified multimodal tactile representations [100]; AnyTouch explicitly addresses static and dynamic information across multiple visuo-tactile sensors [20]; T3 studies transferable Transformers [115]; and UniT uses a unified tactile modelling approach [93]. Sparsh-X broadens the multisensory alignment setting [29], while TactAlign maps human tactile-glove features into a robot tactile space for cross-embodiment transfer [87]. The common scientific question is not simply whether embeddings transfer, but which tactile factors remain identifiable after alignment. A sensor-invariant latent that collapses shear direction or scale may be detrimental to load-path reasoning.

Whole-hand and morphology-aware models extend the unit of representation. HTT uses heterogeneous tactile tokenization and masked learning [5]; FTP-1 incorporates sensor morphology in a large tactile policy [107]; HT-Bench supplies a whole-hand benchmark with egocentric vision [33]. These works motivate explicit metadata for sensor location, normal direction, link identity, and quality. In bimanual systems, morphology-awareness must also prevent a shortcut in which "left" and "right" sensor IDs stand in for task role.

5.3 Temporal tactile representations

Touch is an event stream. Static contact appearance can be ambiguous: the same deformation may correspond to loading or unloading, stable stick or the moment before slip, depending on recent history. TactileVAD highlights geometric aliasing in learned tactile dynamics [64]. Reactive control systems increasingly split slow semantic reasoning from fast tactile reaction; T-Rex uses tactile-reactive control at different time scales [63], while recent slow–fast visuo-tactile diffusion policies pursue a related control principle. ViTacFormer also includes future-touch prediction as part of cross-modal representation learning [26].

HiTac-WAM makes the prediction target more explicit: candidate actions are evaluated through a directed contact $\rightarrow$ deformation $\rightarrow$ slip hierarchy, and persistent forecast error can trigger replanning [96]. VT-MUSE instead learns a frozen sequential visuo-tactile representation using cross-modal temporal alignment, masked-view consistency, and a training-only posterior that sees the full tactile history [92]. These are stronger controls than a single future-frame decoder. They also expose distinct hypotheses: whether physical variables should be ordered explicitly, and whether a deployable prior can retain information learned from richer training-time histories.

Temporal modelling choices include fixed windows, recurrent states, temporal convolution, causal Transformers, and state-space models. Their comparison is often confounded by receptive field, parameter count, and action history. For contact-transition evaluation, causal alignment is essential: a future tactile frame may be a valid training target, but it must not leak into the deployed encoder input. Reporting should distinguish synchronous state estimation, 100–300 ms forecasting, and closed-loop reaction.

Vision/proprioception-only models also strengthen the temporal null hypothesis. TemporalFlow-VLA supervises ordered execution-history queries with training-only robot-surface flow [101]; PredVLA updates a compact recurrent latent

through prediction-error inference [72]. Neither supplies distributed touch, so neither closes the tactile relation gap. They are nevertheless important matched baselines: a new tactile structure should outperform comparable history grounding and prediction-error correction, not merely a memoryless visual policy.

5.4 World models: supervision, conditioning, and feedback

The September frontier separates three often-conflated uses of touch. ME-Dex 1.0 jointly predicts canonical tactile states, video, and actions with engineered hand-region mappings [111]. Yet its observed-versus-zero-current- touch intervention reports nearly identical Random scores (91.92% and 91.74%); its Clean-to-Random comparison also uses zero current touch. With one training seed and qualitative physical deployment, this supports predictive supervision more strongly than a claim that the reported controller needs online tactile feedback. Cross-source reconstruction after pretraining on those sources is also different from held-out-sensor transfer.

DexTouch-WM conditions future tactile prediction on initial sensing and a future pose sequence [68]. Human data improve contact prediction with robot data held fixed, but shared taxel layouts and anatomical pad embeddings supply correspondence. This is an action-conditioned model rather than a standalone observation-only policy; evaluator agreement and synthetic-training utility must be separated because synthetic substitution sometimes worsens real performance. UVTA similarly supplies a useful predictive-supervision comparator [45]: aligned human/robot touch improves real right-hand task scores, but the tactile prediction head is not used for control. Its stage-wise partial-credit scores must not be rewritten as binary success rates, and bilateral hardware must not be mistaken for bimanual experimental evidence.

ContactWorld adds unusually relevant representation controls, including latent-dimension/planning-cost matching, multiple seeds, and same-GelSight hardware comparisons [113]. It supports a conditional rather than universal benefit from touch. Goal observations are drawn from future held-out demonstrations, and real "tactile force fields" are depth/marker-motion proxies rather than calibrated three-dimensional force. Separate hardware models and sensor changes in some comparisons preclude a zero-shot cross-sensor claim. For a contact-flow model, the lesson is to distinguish preserved physical information, predictive objective, goal information, and online feedback necessity before attributing gains to explicit relations.

5.5 Multimodal fusion and spatial anchoring

Tactile measurements acquire meaning from proprioception and vision. Early or late concatenation is a strong baseline because a sufficiently expressive temporal model can learn cross-modal interactions. Cross-attention gives tactile tokens selective access to image, state, or action tokens; ViTacFormer is a prominent recent example [26]. The fair question is whether a proposed structure improves over this baseline at equal observation, compute, and temporal context.

Physical-property fusion adds another alternative. ViTacPhys predicts mass, friction class, and stiffness from human visual–tactile demonstrations and uses the estimates to condition grasping [50]. Such property tokens can explain adaptation across objects without representing a cross-hand load path. A fair system-level comparison should therefore include both property- conditioned fusion and relation-conditioned fusion, and ask which failure modes each resolves.

Spatial anchoring transforms local touch into a shared frame. SaTA anchors tactile awareness using forward kinematics, enabling spatial reasoning beyond sensor index [32]. Object-centric approaches similarly normalize motion with respect to the manipulated entity [10]. Anchoring is particularly valuable for two hands because corresponding tactile features lie in separate kinematic trees. Yet anchoring assumes that the relevant transform is known or perceived; errors in link calibration and object pose should be tested, not hidden.

The early-October evidence sharpens spatial and predictive controls further. SimpleTouch combines a frozen pretrained tactile encoder with task-specific VLA/expert learning and removes the future-touch branch at deployment [97]. "Without tactile-policy pretraining" still uses upstream encoder/VLA pretraining. Its main insertion comparison also differs in observation refresh (25 versus 50 executed actions); matched-horizon ablations are the cleaner representation comparison. VISTA supplies a pose-conditioned spherical-fusion alternative [88]. Its equivariance is defined for selected finite groups and joint workspace rotations, with current end-effector orientation observed. Single-seed, single-gripper evidence and differing sensor/ group settings across broad baselines limit stronger transfer claims. Both works make execution horizon and observation refresh part of the matching budget, alongside local encoder, data, history, and parameter count.

5.6 Direct cross-hand completion evidence

BiView-Touch changes the boundary of the representation gap: it learns bilateral human tactile representations by geometry-conditioned cross-hand completion, with static anatomical-region centroids and temporal/layout counterfactuals [46]. Its matched temporal, within-hand, and dense-fusion comparisons, together with donor-context substitutions and region permutations, provide direct evidence that contralateral context can be useful. Dense fusion and cross-attention perform similarly in the principal readout comparison, so this does not establish architectural superiority of geometry-conditioned attention. Cross-hand representation learning itself should therefore no longer be presented as absent.

Table 4: Representation families, what they organize, and the unresolved cross-interface question. Entries are representative rather than exhaustive.

Family	Representative works	Primary organization	Unresolved bimanual question
Local SSL	T-Dex, Sparsh [23, 28]	contact appearance, geometry, dynamics	Does the latent retain load magnitude/direction and interface identity?
Cross-sensor alignment	UniTouch, AnyTouch, T3, Sparsh-X, TactAlign [20, 29, 87, 100, 115]	shared tactile factors across hardware	Which physically relevant factors are lost to invariance?
Whole-hand / morphology-aware	HTT, HT-Bench, FTP-1 [5, 33, 107]	sensor layout, link/body coverage	Can roles and topology transfer across two heterogeneous hands?
Temporal dynamics	TactileVAD, T-Rex, HiTac-WAM [63, 64, 96]	event history, directed tactile state, and action-conditioned change	Are transitions predicted causally, and does the directed hierarchy beat a matched flat head?
Sequential cross-modal	ViTacFormer, VT-MUSE [26, 92]	vision–touch–state interaction and training/deployment temporal latents	Does explicit interface structure add value to matched sequential fusion?
Physical-property latent	ViTacPhys [50]	mass, friction, and stiffness estimates	Do property tokens explain adaptation without cross-interface relations?
Spatial anchoring	SaTA [32]	tactile features in kinematic space	How robust is the representation to pose error and topology change?

The evidence is nevertheless narrower than robotic load-path identification. The experiments are offline human sensing; several phase/contact labels derive from the same tactile input, whereas wrist-motion labels come from independent tracking. No closed-loop robot study is reported. The held-out buckle task tests task transfer, but its handover/connection motifs occur elsewhere in training; the split does not hold out a declared participant/edge topology. Static anatomy, predictive dependence, object-mediated force transmission, and calibrated load inference remain distinct claims. A robotic study should retain the cross-hand completion and dense-fusion controls while adding independent interface loads, participant/session holds-out, and repeated pretraining seeds.

5.7 What existing objectives supervise

Pretraining objectives privilege different information. Reconstruction preserves pixel detail but may waste capacity on illumination or marker appearance. Masked prediction encourages context use but can exploit spatial redundancy. Contrastive alignment emphasizes shared content but can suppress modality-specific physical signals. Future prediction encodes dynamics but may average multimodal outcomes. Task reward discovers control-relevant features but can exploit spurious state or demonstration regularities. No objective is universally correct; its sufficiency must be judged against the intended quantities and topology.

For bimanual coordination, promising auxiliary targets include contact onset and release, future interface occupancy, load-share ratio, future local wrench, slip risk, and calibrated uncertainty. These are not merely extra labels. They test whether the latent exposes a mechanism that a downstream policy could use. Where labels are available only in simulation, privileged-to-observation distillation can train an observable student, but evaluation must verify that the student does not require privileged inputs at test time.

5.8 Necessary baselines for representation claims

A new structured encoder should be compared against more than raw concatenation. At minimum, a matched study should include: proprioception-only; tactile-only; concatenation with the same local encoder; cross-attention; a temporal Transformer or recurrent model; a prediction-error recurrent model; a flat multi-head and a directed tactile hierarchy; training-only temporal grounding; spatial anchoring where poses are available; a static graph; the proposed inferred structure; and an oracle structure used only as an upper bound. Parameter count, input history, optimization budget, augmentations, and pretraining data must be controlled or reported. Without these controls, an improvement may come from capacity, temporal context, objective ordering, or privileged supervision rather than representation structure.

5.9 Canonicalization and scale: the 14 October evidence

TACROSS and TouchScale strengthen human-to-robot learning baselines but require separate interpretations of alignment, scale, and physical supervision [6, 42]. TACROSS uses anatomical contact fields, sensor-specific adapters, validity masks, and causal inter-finger/temporal encoding. Its intensity and instability channels are relative proxies, not independently calibrated force or slip. Human-input ablations do not establish the necessity of online robot touch, and its four-task robot evaluation is single-hand rather than cooperative bimanual control.

TouchScale's cross-sensor evaluation pools native sensing into twelve anatomical regions and uses sequence-normalized pressure. Its approximately size-matched comparison must be distinguished from full-scale training gains. Action-free future-touch mid-training improves subsequent single-hand robot control, but matching robot post-training data does not match total pretraining data or compute. Bimanual human collection is not evidence of bimanual robot execution. Artifact claims also require dating: the official dataset card on 14 October advertises 100 hours available, with the full 500-hour release planned and access gated; TACROSS's project resources remain planned. Neither a paper's collection size nor a repository URL establishes unrestricted, complete artifact availability.

PACE provides a controlled temporal-organization baseline: smoothed contact-state conditioning, action residuals from contact changes, and training-only future contact prediction [12]. It reports 136/150 real insertion successes versus 69/150 for tactile concatenation with common demonstrations and control frequency. However, Appendix A.6 reruns the full ACT policy each control step; this is not evidence of a separately asynchronous intra-chunk reflex. Its alternative feedback-signal experiments do not establish held-out-sensor transfer, and task/geometry-specific training must remain explicit.

5.10 Predictive representations: specify what closes the loop

CTP, ForceWAM, and ForeTac expose three different routes from contact prediction to action [11, 61, 95]. CTP queries ordered motion/wrench/deformation tokens from a target-object demonstration using current sensing; a reference is a behavior specification, not a deployable observation of the future. ForceWAM conditions execution on generated visual/wrench futures and causal measured wrench feedback; adding exploratory play improves execution while also increasing total experience. ForeTac uses predicted tactile latents to guide candidate actions online, but adds auxiliary training and task-specific quality supervision. Its 20 October v2 replaces the four-class scorer with binary good/bad prototypes and a direct logit margin, while reporting unchanged headline outcomes; the revision no longer specifies seed 42. Matching a frozen base-policy checkpoint does not equalize the entire guidance system's training or computation. None of these single-arm results alone establishes bimanual contact-topology transfer.

ILA changes the action target itself to a joint motion–tactile latent [117]. Tactile-latent supervision can help even when policy input touch remains fixed across ablations. However, complete recorded chunks train the encoder, whereas deployment starts from current observations and executes 128 actions before replanning. Offline realized tactile outcomes used for reward or probes are not online future measurements. This makes prediction, latent supervision, and actual feedback cadence distinct axes of comparison.

DeTac further illustrates supervision accounting at the local sensor level [112]. Separating marker and shading signals using real aligned pairs improves downstream estimation, but the large comparison with synthetic- overlay training is not an architecture-only effect. Its matched paired-data U-Net is a more diagnostic control. Qualitative cross-sensor examples and local force/geometry probes do not establish closed-loop bimanual transfer.

5.11 Information roles in tactile prompting and pretraining

Frozen multimodal agents make tactile information packaging a meaningful baseline, but they do not eliminate demonstrations, reference engineering or latency constraints [83, 86, 98]. GPT-as-Tactile-Policy combines online descriptors with reference skills distilled from demonstrations; its 93.8% simulator mean is below N0-VTLA's 96.2%, despite the title's broad superiority claim. Its tactile ablation removes both online information and tactile references. By contrast, Tactile-ETA holds current touch constant and changes demonstration touch: the main simulator comparison therefore tests contextual information, not sensor necessity. Its physical comparison adds a complete multimodal demonstration and uses only five trials per condition per task. TacTalent's paired modality tests include tasks that worsen with touch and a relaxed benchmark success rule. Approximately fourteen-second API latency and 60 Hz command execution describe different clocks. These studies motivate matched reference, selection, history, criterion and feedback-budget controls, not universal hardware-policy rankings.

HuTact and Spark-0 separate useful initialization from online sensory dependence [8, 106]. HuTact's human pretraining uses mesh-distance pseudo-contact, not measured human force; target tokenizers and policies require adaptation. Spark-0 offers a stronger matched-RGB/compute pretraining contrast: real success rises from 44% to 52% when pretraining retains touch, while both downstream variants already use touch. This isolates pretraining information rather than runtime tactile necessity. Target-embodiment fine-tuning and a common glove family do not establish zero-shot morphology or cross-sensor transfer.

Future observations must also be distinguished from prospective commands. ACTS conditions offline world-model prediction on the entire recorded motion and tactile-derived force window [53]. It supplies useful trajectory-conditioned human-to-robot prediction evidence, not causal autonomous planning from currently available sensing. Tactile persistence beats it on PSNR for five of six robot tactile streams, while LPIPS favors it on five; static appearance, event timing and physical prediction should not be conflated. Ego2Contact learns anatomically aligned pressure fields, but its sensor-substitution claim is confined to simulation [103]. The real policy uses predicted touch for offline post-training and measured glove touch at deployment; substantial paired target adaptation precedes transfer. Neither future-conditioned prediction nor adapted visual contact completion alone validates independent cross-interface force observability.

5.12 Runtime touch, local control, and future targets

Compact sensing remains a serious comparator. Blind Baoding obtains sustained vision-free two-ball rotation from four frames of proprioception and sixteen binary contact channels [18]. Removing touch from the same checkpoint reduces mean real rotations from 34.75 to 5.00 in ten trials per configuration. This supports runtime tactile dependence, but the separately trained proprioceptive baseline has a different slew setting. Neither the cumulative transfer recipe nor one fixed ball pair establishes a rich representation advantage or topology generalization.

ContactPolicy makes local temporal organization relevant to actual bimanual control, rather than merely to offline representation probes [3]. Per-arm contact policies share intent routing; five of six hardware tasks are bimanual. Tactile–force fusion improves average success, but its 60 Hz local controller is compared with 30 Hz baselines. Representation, local control frequency, shared task intent and modality access therefore require separate matched controls. This is positive bimanual evidence, not an independently labeled cross-interface load model.

6 Graphs, Contact Fields, and Physical Grounding

6.1 A graph is a hypothesis about factorization

Interaction Networks established a general relational inductive bias: entities and their pairwise relations can be modelled separately, with messages aggregated to update entity states [4]. NerveNet applies graph policies over articulated morphology [81]; graph simulators learn physical dynamics over interacting particles or bodies [71]; Graphormer demonstrates that Transformer attention can be modified with graph-structural encodings [105]; and AnyMorph studies policies across morphologies [77]. Together these works make “uses a GNN” an architectural choice, not a sufficient novelty claim.

For tactile bimanual manipulation, a graph encodes a stronger hypothesis: task dynamics factorize around interfaces and the paths connecting them. This hypothesis is useful only if (i) node and edge semantics correspond to deployable or inferable quantities, (ii) the factorization improves sample efficiency, robustness, interpretability, or intervention response, and (iii) the same gain cannot be explained by additional state or parameters.

6.2 Taxel, finger, and morphology graphs

Graph construction at the tactile-array or finger level is relatively direct. Distributed tactile sensors have known positions on an articulated hand; convolution or message passing can share information along this geometry. Multi-finger in-hand manipulation with distributed tactile GCNs demonstrates this idea [21]. TacGNN uses hierarchical graph structure for blind tactile-based in-hand manipulation [102]. TactiGraph treats event-based tactile measurements asynchronously for contact-angle prediction [70], and TouchWGNN develops spatio-temporal tactile graphs for multimodal dexterous manipulation [62].

These models address meaningful spatial and temporal organization. However, a taxel adjacency graph is not the same as a dynamic task contact graph. Taxels can remain neighbours in the sensor layout whether or not their links contact the same object; fingers can be kinematically adjacent while transmitting loads through different entities. Bimanual load-path work therefore needs at least two levels: relatively stable morphology/sensor structure and dynamic inter-entity interfaces.

DexGraspFlow sharpens the distinction between kinematic and contact topology [118]. Its link tokens combine URDF shortest-path biases with relative geometry; unseen finger-removal variants genuinely alter the kinematic structure. This is meaningful morphology generalization, but it does not hold out dynamic contact participants or infer cross-hand load paths. Real grasps use separately trained hand demonstrations, pose estimation, inverse kinematics, and closure offsets, limiting a zero-shot embodiment claim. The implication is to evaluate morphology and contact-graph transfer separately, not dismiss either by conflating their definitions.

6.3 Dynamics graphs and contact discontinuities

RoboPack learns tactile-informed recurrent graph dynamics for dense packing and uses the model for planning/control [1]. Its relational factorization is compelling because tool, objects, and environment interact over time. ContactNets targets

discontinuous contact dynamics with a structured implicit model [67]; graph network simulators have been shown to model rigid contact transitions [2]. This line of work suggests two design principles for bimanual representation: predict changes at interfaces, and avoid treating discontinuity as ordinary smooth motion noise.

MeshPriorDiT adds a useful hierarchical counterexample to an architecture-only comparison. An action-conditioned mesh GNN supplies topology and grasp-constraint priors, while a flow-matching diffusion Transformer models residual and global cloth effects [84]. Its evidence concerns deformable mesh dynamics rather than tactile load transfer, but the factorization is directly relevant: structured local priors and expressive global residuals can be complementary. A bimanual study should therefore compare graph-only, global-residual-only, and combined models rather than forcing a false GNN-versus- Transformer dichotomy.

Yet dynamics prediction can be correct for the wrong reason. Exact object pose, collision geometry, and contact labels can make a simulator graph accurate while an observation-only system remains underdetermined. A useful paper must specify which signals construct the graph at training and at deployment. "Physics grounded" should describe targets, conservation constraints, frames, or interventions—not merely that the task occurs in a physics engine.

Dex-X makes the distinction between removing privileged state and removing reference dependence concrete [7]. Its spatialized tactile-force student drops current object-pose and geometric privileged blocks but retains next-step motion references and the demonstration's target object pose. Its bimanual handover result belongs to simulation; real evaluation includes single-hand manipulation with strongly mixed object-generalization outcomes. This is a useful spatialized-force and distillation comparator, but should not be called reference-free, real bimanual load-path inference, or topology transfer. CoRL acceptance was author-reported at this audit, not independently confirmed from an official venue record.

CQDC-Warp supplies a complementary mechanics comparator: compliant contact and geometry derivatives support local MPC, including simulated shared-object lifting [19]. Realized contacts are optimized without a prescribed mode schedule, but scene models, initialization, and intended contacting links remain inputs. Hardware evidence is single-hand trajectory replay, not closed-loop bimanual tactile inference. Its GPU-batched versus CPU-sequential timing comparison also prevents attributing the full speedup to the contact formulation.

6.4 Contact fields, modes, and anchors

Graphs are not the only structured representation. Neural Contact Fields track extrinsic contact through tactile sensing [27]; object-centric contact-field approaches extend this spatial view to visuo-tactile state estimation and tool interaction. Semantic Contact Fields predict semantically meaningful contact/force structure for tool manipulation [52]. Contact-mode control represents discrete constraint regimes for non-prehensile manipulation [65]. Contact-anchored policies and contact-grounded policy learning similarly use contact as a privileged or learned organizing variable [17, 94].

Recent work also clarifies two meanings of "contact structure." One-shot learning by identifying physical interactions extracts making/breaking contact events from a demonstration and uses them as transition conditions for hybrid position–force control [66]. CoToGrasp conditions grasp synthesis on a requested contact topology in a canonical workspace [59]. Both make structure explicit, but the former uses a demonstrated event state machine and the latter a prescribed grasp-contact layout. Neither is equivalent to inferring a time-varying cross-hand topology from raw deployment observations. They are valuable controls precisely because they test how much structure can be specified rather than learned.

CADeT is positive evidence that useful interaction structure need not be supplied as simulator truth [30]. It infers a belief over predefined coupled, decoupled, and blocked tissue-transmission modes from visually tracked responses and physical probes. In 150 simulation trials with mode truth withheld, it identifies 149 modes correctly; matched information-guided probes require fewer additional actions than random probes. Phantom and ex-vivo experiments extend the evidence beyond simulation. This is visual, fixed-family mode inference, with nominal calibration before blockage, rather than tactile cross-hand graph inference or held-out-participant topology transfer. GP covariance informs mode likelihoods but is not propagated through the MPC horizon; entropy convergence is not probability-calibration evidence, and contact force/strain safety is not measured or guaranteed.

Fields are attractive when contact location varies continuously over object surfaces; discrete graphs are attractive when entity/interface identity and message routing matter. Hybrid representations can use a field to infer contact location and confidence, then instantiate an entity-level relation. The decisive comparison should be empirical: a field, graph, anchored token set, and unstructured attention model should receive the same observations and predictive targets.

6.5 PhysGraph as a boundary case

PhysGraph is a directly relevant early graph-policy formulation for bimanual hand–tool–object manipulation [44]. It organizes hand, tool, and object information in a graph Transformer and reports strong simulated control. Its input regime, however, includes simulator state, fingertip resultant forces, object/tool geometric encodings, and future reference. It should therefore be read as evidence for relational policy factorization under privileged state, not as tactile-array representation pretraining or observation-grounded contact inference.

Table 5: Structured-representation choices. The final column states what must be demonstrated rather than assumed.

Structure	Natural unit	Typical source	Validity test
Sensor/morphology graph	taxel, link, joint	calibrated layout and kinematics	robustness across layouts, missing sensors, and embodiments
Static task graph	hand, object, tool	designer-specified entity relations	advantage over matched attention without dynamic edge changes
Dynamic contact graph	active interface	tactile/proprioceptive inference or simulator contact	inferred–oracle gap, transition timing, false-edge interventions
Object-centric anchor	local feature in object frame	visual/kinematic pose estimate	degradation under pose error and unseen grasps
Contact field	continuous surface point/force	tactile/vision and geometry	contact localization, force consistency, surface/OOD generalization
Contact mode	discrete constraint regime	estimator, demonstration, or privileged label	mode-transition accuracy and closed-loop recovery
Physics residual/constraint	force balance, momentum, complementarity	measured or simulated quantities	calibration, counterfactual consistency, no label circularity

A local audit of the public main-branch snapshot at the corpus freeze found that the bias generator exposes contact-mask and geometric-position inputs, while the default forward call does not pass them; the resulting public path uses the configured topology rather than the advertised dynamic contact/geometric bias. This is a reproducibility observation about that code snapshot, not evidence that the authors' reported experiments used the same path. The correct experimental response is to reproduce both the stock public path and a clearly documented corrected-bias path, then avoid attributing their difference to the paper without author confirmation.

This case illustrates why code-level observability matters. A diagram can show dynamic edges while the executed model uses static connectivity; a paper can describe tactile feedback while the input is a low-dimensional simulator force; and a physically plausible target can be constructed from unavailable future state. Reproduction should trace every tensor from sensor or simulator source to the model and record its deployment availability.

6.6 Physical objectives for cross-interface structure

Suppose an inferred interface edge $e_{ij,t}$ has probability $p_{ij,t}$ and an estimated wrench $\hat{\boldsymbol{w}}_{ij,t}$ expressed in a common object frame. Useful objectives can combine:

$$\mathcal{L} = \lambda_e \mathcal{L}_{\text{edge}} + \lambda_w \mathcal{L}_{\text{future-wrench}} + \lambda_s \mathcal{L}_{\text{slip}} + \lambda_p \mathcal{L}_{\text{phase}} + \lambda_c \mathcal{L}_{\text{consistency}}. \tag{1}$$

The consistency term might penalize disagreement between predicted interface wrenches and measured object acceleration or enforce action–reaction relations when both sides of an interface are observed. It should be soft and uncertainty-aware: unmodelled compliance, gravity-compensation error, and unobserved environmental contact make exact balance inappropriate. The model should also predict confidence, because a missing edge under sensor occlusion is not equivalent to a confidently absent contact.

For sustained contact, the state should additionally represent slow relaxation. Temperature-coupled viscoelastic force estimation shows that identical-looking instantaneous configurations can correspond to different force histories [82]. Interface objectives should therefore be evaluated at multiple contact durations and, where relevant, condition on temperature or an identified relaxation state rather than treating drift as independent noise.

The central empirical question remains comparative. If a temporal Transformer with the same inputs predicts future wrench and phase equally well, explicit edges may add interpretability but not predictive value. If structure improves data efficiency, transition accuracy, or transfer to a new topology—and the benefit disappears under random or time-shuffled edges—the relational hypothesis receives stronger support.

6.7 Local structure is not cross-interface load structure

Two October studies sharpen this distinction. Minimal optical-flow features on a fixed tactile grid support conditional rotation-direction classification under simulated gravity variation [57]. Mixed-gravity training covers seen domains; it is not held-out-gravity transfer. Excluding ambiguous rotation and linear-slip events also limits safety interpretation, while classifier-only latency omits optical-flow preprocessing. There is no real sensor or closed-loop control test.

A finite-element-supervised GNN uses seven marker nodes and eleven fixed edges for sensor-local contact localization/deformation [76]. Its 82 FEM-validation configurations must be distinguished from 400 additional real GNN-validation samples. The absence of a matched non-graph or shuffled-edge control prevents attributing accuracy specifically to graph structure. Its MuJoCo integration disables skin–indenter collision and is visualization rather than physically coupled contact dynamics. Both studies are useful local controls; neither validates inferred bimanual load topology.

TacSpin's depth/shear geometry separates rotation axis and magnitude and combines opposing fingertip responses [90]. Calibration/evaluation tags are not required at deployment, but mixed-axis sphere results do not preserve all fixed-axis gains: removing circulation can improve trajectory/end-point metrics. A useful representation review must retain that negative result rather than promote local geometry to universally robust contact structure.

The edge-native temporal–graph dual expert is an even sharper attribution example [116]. Its fixed finger graph contributes to instance classification, while all learned Stop predictions come from the temporal expert. Angle/load proxies are not direct tactile arrays; five labels each represent one physical instance. A trajectory split is therefore within-instance recognition, and threshold-teacher Stop labels are not independent grasp-stability truth. The larger fused model cannot establish a graph-caused control benefit.

MorphoPlanner broadens the structural comparison to distributed body points, force/proximity sensing, fixed body connectivity and attention [80]. Its large training set is simulator/planner- supervised, the 98/1/1 split unit is unspecified, and a separate controller executes predicted morphology. The 95.5–100% headline concerns simulated movable-obstacle reaching, not real dual-arm success. Stating a hardware trial protocol without a numerical result does not justify that translation. Without matched graph/attention ablations, obstacle-density variation is not proof of contact-edge transfer or inter-arm load inference.

7 Bimanual Coordination, Handover, and Cooperative Tool Use

7.1 Bimanual learning is broader than tactile coordination

Bimanual manipulation has grown rapidly through simulation benchmarks, retargeting, imitation, and scalable demonstration generation. Bi-DexHands provides a broad simulated reinforcement-learning benchmark [9]. Bimanual Dexterity for Complex Tasks and related real systems demonstrate increasingly capable coordinated policies [74]. DexMimicGen generates bimanual data from demonstrations [36]; ManipTrans transfers behaviours through residual learning [43]; object-centric human-motion approaches organize demonstrations around manipulated objects [10]. These works establish task and data scalability. They do not automatically test whether a tactile latent explains cross-hand forces or contact transitions.

Late-August work further raises the vision/proprioception-only baseline. Training-time modality masking improves robustness of a query-based bimanual VLA without adding a tactile relation model [13]; PartialBiGrasp infers hidden local geometry from partial point clouds and searches for force-closure-compatible grasp pairs [39]. CLAP learns a cross-embodiment action-conditioned video world model with bimanual data among its supported settings [48], while rapid on-robot juggling learns local dynamic models around a global prior under a reachable safe-set constraint [41]. Together they explain more bimanual success through visual geometry, robust fusion, video prediction, and rapid dynamics adaptation. A tactile representation must therefore demonstrate an incremental benefit under occlusion, friction change, slip, or simultaneous load transfer, not rely on bimanual task success alone.

DemoMimic strengthens the geometry-only alternative: surface-normal alignment and sustained-contact rewards train a privileged residual teacher, then a visually guided policy deploys with wrist depth and proprioception [73]. Its real and simulated comparisons show why near-ceiling simulation can hide real contact loss. Evaluation spans objects and two hand embodiments, but uses task-specific policies and graded articulation, distance, or stage scores rather than one uniform binary success metric. This motivates geometry/alignment controls without establishing tactile benefit or held-out contact-topology transfer.

The new surveillance evidence makes the baseline question more specific. GRASP grounds bimanual visual policies through training-only forward-kinematic and inter-arm pose losses plus visual teacher supervision [69]. Matched training budgets and real tool-use gains make this a substantive non-tactile comparator. Yet the full method performs worse than its query-only ablation under randomized simulation, so added physical supervision does not uniformly improve robustness. Kinematic accuracy should therefore be tested alongside, rather than substituted for, contact-force evidence.

Physical Canvas moves the tactile comparison from policy inputs to imagined bimanual rollouts [91]. Its shared visual–tactile canvas covers handover, reorientation, insertion, and cap manipulation; task-state supervision improves policy-outcome correlation, and imagined reinforcement learning improves subsequent real success. These results support outcome-aligned tactile world modelling, with real observations initializing generated futures. They do not establish calibrated force dynamics or identify object-mediated load paths. Moreover, modality-consistency training slightly reduces event F1 in one ablation. A useful comparison must therefore report event fidelity and real control benefit separately from video quality or aggregate outcome correlation. Neither study establishes strict contact-topology transfer, and these surveillance additions do not alter the historical zero-of-30 core result.

This distinction matters for literature positioning. The broad claim "bimanual manipulation is underexplored" is no longer defensible. The narrower intersection of *observation-grounded tactile structure*, *explicit interface/load targets*, and *matched representation comparisons* remains much less populated in the mapped corpus. A contribution should therefore be framed around that intersection, not around the existence of two hands.

Bi-Touch is an important early direct baseline: it used a concatenated tactile policy state and sim-to-real reinforcement learning for contact-rich bimanual pushing, reorientation, and gathering tasks [47]. It shifted the question from whether touch can help bimanual control to whether an explicit relational representation adds value beyond a matched tactile policy input. A new graph method should therefore include an equivalent concatenation baseline even if the original Bi-Touch implementation is not the primary reproduction target.

7.2 Roles, symmetry, and mechanical necessity

A bimanual taxonomy can distinguish symmetric and asymmetric actions, shared and separate objects, coordinated and independent motions, and temporal role changes [40]. For representation research, two additional questions are decisive:

1. **Is coordination mechanically necessary?** If one hand can complete the task while the other supplies a decorative or easily ignored motion, task success does not diagnose cross-hand representation.
2. **Can the roles be exchanged?** A representation that encodes fixed left/right identities may fail when actor and stabilizer swap, even though the physical relation is unchanged.

BUDS makes stabilizer–actor coordination explicit and demonstrates that one hand can create conditions for the other [22]. This is a useful conceptual bridge from shared-object manipulation to tool–workpiece tasks. However, role assignment should be coupled to measurable necessity: removing, delaying, or perturbing the stabilizing hand ought to change forces, error, or success in a predictable way.

7.3 Handover as a load-transfer benchmark

Handover provides an unusually clean phase structure. Human-inspired and robot–robot/robot–human controllers have long used load, pull, grip force, or slip cues to govern transfer and release [15]. Contact-aware dexterous handover further emphasizes when and where contacts occur [85]. VTDexManip supplies a reproducible simulated bimanual handover task with visual-tactile observations [49]; recent tactile-reactive systems expand the scale and realism of bimanual touch [63].

Temporal tactile encoding with compliance is now a direct real-system comparator. Marra et al. compress approximately 0.9 s of fingertip-force history into the same 24-dimensional size as the instantaneous force vector and deploy it with vision, proprioception, and compliant control on a two-handed ergoCub box handover [56]. In 100 trials per policy with ten new participants, history plus compliance achieves 93% expected release-or-hold outcomes, versus 54% without tactile policy input (force-based compliance retained) and 82% without compliance. The outcome includes correctly continuing to hold; it is not a transfer-only success rate. This establishes a strong causal-history/compliance baseline without an explicit cross-interface graph. Separate compliance datasets also limit a fully factorial interpretation. Misleading pulls, wrong-direction forces, no-pull grasps, and self-motion should be tested before attributing safe release to graph structure.

The most informative handover interval is dual contact. Let $F_L(t)$ and $F_R(t)$ be gravity-axis support estimates expressed in a common object frame. A simple load-share target is

$$\rho_R(t) = \frac{\max(0, F_R(t))}{\max(0, F_L(t)) + \max(0, F_R(t)) + \epsilon}. \tag{2}$$

This ratio is interpretable and less scale-sensitive than absolute force, but it is only valid if the force estimates are independently calibrated and frame aligned. Contact phase, time-to-support, future receiver wrench, giver slip risk, and release readiness provide complementary targets. The point is not to impose one scalar definition; it is to make the transfer mechanism measurable.

Dynamic throw–catch handover explores timing, ballistic motion, and impact recovery [31]. It should be a later robustness task rather than the primary cross-hand load benchmark, because the flight phase removes the simultaneous object-mediated path. A quasi-static transfer with a speed perturbation, object-mass change, or receiver delay offers cleaner causal tests first.

Robot juggling reinforces this sequencing: impressive rapid online adaptation in a dynamic bimanual task can be achieved from onboard vision and a local dynamics model [41]. It is therefore a strong adaptation reference, but its flight/contact alternation does not provide the persistent, simultaneous object-mediated load path needed to validate cross-hand tactile structure.

Table 6: How the two principal task families complement each other. "Mapped evidence density" is a qualitative inference from the curated corpus, not a bibliometric claim.

	Quasi-static handover	**Cooperative tool–workpiece use**
Core topology	hand–object–hand, changing support	hand–tool–workpiece–hand, persistent role asymmetry
Primary signal	load redistribution and release	working-contact wrench, stabilizer reaction, slip
Main advantage	clean phases and interpretable transition	harder multi-hop transfer and practical task relevance
Main confound	object pose/motion can reveal phase	visual trajectory, periodic motion, and compliance can reveal task state
Mapped evidence density	established handover/control literature; few matched representation studies	established tool/contact systems; few matched cross-hand load-flow studies
Best research role	first controlled representation benchmark	gated second topology after mechanism is established

7.4 Cooperative tool–workpiece manipulation

One hand holding a brush while the other holds a plate is closely aligned with the same representation problem, but its topology is more demanding. The actor hand senses its grasp on the tool; the stabilizer hand senses the workpiece; the working interface between brush and plate is remote from both local grasp interfaces. Force and motion propagate through at least three contacts. This task therefore tests whether a representation can move from $H_L \leftrightarrow O \leftrightarrow H_R$ to $H_L \leftrightarrow T \leftrightarrow W \leftrightarrow H_R$.

The adjacent literature supplies components rather than a saturated direct benchmark. Tactile Tool Manipulation learns tool-related control with tactile feedback [75]; compliant tool–environment dynamics and extrinsic contact-mode control model the working interface [65, 79]; RoboPack demonstrates tactile-informed relational dynamics in tool-mediated dense contact [1]; TACO covers tool–action–object semantics [51]; TAMEn targets tactile-aware closed-loop data collection in contact-rich tasks such as dish washing [89]; and Semantic Contact Fields make tool contacts explicit [52]. These works make the area adjacent, not empty. What remains scarce is a controlled study of *cross-hand tactile load propagation* with a held workpiece and matched representation baselines. TAMEn is a direct system competitor for a dish-washing demonstration; novelty cannot rest on performing that task with touch, but must come from observation-grounded load-path inference, mechanism-level evidence, or cross-topology transfer.

7.5 Designing a diagnostic tool task

A diagnostic task should make the second hand necessary while remaining simple enough to label. A rigid stylus or wiping puck against a plate is preferable to a deformable brush for an initial experiment: the contact location, normal, and applied path can be controlled, and simulation is more credible. The task can then progress to a compliant brush once sensing and analysis are stable.

Useful experimental factors include tool stiffness, workpiece mass, friction, path curvature, contact-force target, stabilizer grasp, actor/stabilizer role swap, and brief perturbations of either hand. A *coordination-necessity check* should compare two hands, fixed workpiece, passive second hand, and removed second-hand tactile input. If the actor succeeds with no measurable cost when the stabilizer is removed or frozen, the task is not a strong cross-hand benchmark.

The tool task should enter a research programme after the simpler handover study has established three points: inferred structure improves at least one transition/load target over matched unstructured baselines; the gain is not due to privileged state; and structure remains useful under object/grasp holds-out. Before those conditions, tool use is valuable as a frozen task specification and offline proxy, but a full hardware campaign risks combining unsolved sensing, calibration, control, and representation problems.

7.6 Cross-topology transfer as the stronger claim

Training and testing separately on handover and tool use would produce two task results. A stronger representation claim asks whether a model pretrained to infer interfaces and loads on handover transfers to the longer tool topology with fewer labels or demonstrations. The test should distinguish:

- **encoder transfer:** freeze the local tactile encoder and retrain only the relation head;
- **relation transfer:** preserve edge functions while instantiating a new entity topology;

- **full fine-tuning:** initialize all weights and measure data efficiency;
- **zero-shot probing:** test whether phase/load variables are linearly recoverable without task fine-tuning.

If only full fine-tuning helps, the model may provide generic initialization rather than compositional topology transfer. If relation modules transfer under role swap and added tool/workpiece nodes, the evidence for structured factorization is substantially stronger.

7.7 Tracker-assisted adaptation as a non-tactile control

Bimanual RMA supplies another relevant comparison: a shared latent inferred from history and wrist depth can adapt coordinated manipulation without an explicit tactile graph [38]. Its deployment observations still include externally tracked object position. Training-only contact impulses and contact rewards must not be recoded as online tactile-array sensing. Four-seed simulation results are complemented by 14 successful lifts in 20 real trials of one box; there is no hardware baseline or held-out contact-topology test. A same-recipe PPO comparison does not isolate the value of structure from latent history and adaptation. The result strengthens the case for system-identification controls, without validating independent cross-hand load allocation.

7.8 Stage routing and compliance are distinct explanations

DEFT-VLA adds real bimanual pipette aspiration to the force-aware control evidence [34]. Its contact expert uses wrist-wrench history and stage-localized modality masking, while the base expert retains other phases. Pipette routing is learned from causal observations at deployment, but whiteboard wiping uses manually triggered switching. Contact-stage success is conditional on reaching that stage, and aspiration-volume error is reported on successful trials. These denominators, routing assistance, and expert capacity must remain visible when comparing a structured tactile representation. Held-out grasp heights are not a new contact-participant topology.

FOMO-Dex instead configures force-responsive directions in a task frame and compares selective with isotropic compliance [108]. This is single-hand tool/control evidence, not bimanual load inference. Its corrected- action imitation policy deploys with a stiff controller, so improved learned actions must be distinguished from online compliance. Force thresholds are operational limits rather than calibrated safety guarantees; force reduction, turns completed, and binary task success remain separate outcomes. Together, these studies strengthen stage-aware and compliance baselines without proving that an inferred cross-hand graph supplies the causal benefit.

ILA's real plate-holding and sponge-wiping setup supplies direct cooperative bimanual tool evidence [117]. It should strengthen the comparison set, not be dismissed because its representation is not a graph. Yet fixed chunk-start sensing, long open-loop execution, sensor-native deformation proxies, and mixed cut/rollout success units prevent treating its task scores as independent validation of cross-hand physical load propagation. Target-task success and a mechanism-specific representation claim remain different tests.

8 Evaluation: From Task Success to Representation Evidence

8.1 Why task success is insufficient

Closed-loop success is necessary for a control method but weak evidence about a specific latent mechanism. A policy can infer phase from vision, memorize object trajectories, exploit a scripted receiver, or rely on privileged object state. It may succeed without using tactile inputs at all. Conversely, a representation can predict contact transitions accurately while a poorly tuned controller masks the benefit. Evaluation should therefore connect four levels: observation integrity, representation probes, open-loop dynamics, and closed-loop behaviour.

8.2 Matched-budget baseline matrix

The baseline matrix in table 7 isolates the source of an improvement. All learned variants should share the same local tactile encoder, observations, history, training examples, optimizer budget, and approximately matched parameter/compute count. If a graph variant receives link poses while a Transformer does not, the study tests information access, not factorization. Two additional controls are necessary after the core-paper audit. A predictive contact-grounding baseline tests whether future tactile/state objectives alone explain the gain [26, 94]; a dynamic taxel graph tests whether local sensor-level message passing already suffices [62]. The surveillance update adds three more controls: a directed contact–deformation–slip predictor [96], training-only temporal-flow grounding [101], and prediction-error recurrent inference [72]. Cross-embodiment visual prediction such as CLAP is a useful system-level baseline where video pretraining is available [48], but it must receive the same deployment observations when used to isolate the value of touch.

Table 7: Minimum matched-budget experiment matrix. The oracle graph is an upper bound and must never be presented as a deployable method.

Model	Same local encoder	Temporal context	Explicit structure	Question isolated
Proprioception-only	n/a	matched	none	Is touch needed?
Tactile concat	✓	matched	none	Does local touch help?
Cross-attention	✓	matched	implicit	Is flexible token interaction sufficient?
Temporal Transformer	✓	matched	implicit	Is time/context the main source of gain?
Prediction-error recurrent	✓	matched	implicit/predictive	Does observation-driven latent correction explain the gain?
Predictive grounding	✓	matched	implicit/predictive	Do future tactile/state objectives explain the gain?
Directed tactile hierarchy	✓	matched	ordered physical heads	Does contact $\rightarrow$ deformation $\rightarrow$ slip ordering beat flat heads?
Training-only temporal grounding	✓	matched	privileged supervision only	Does richer training supervision, rather than deployed structure, explain the gain?
Spatially anchored tokens	✓	matched	coordinates	Does a shared frame suffice?
Dynamic taxel graph	✓	matched	local learned edges	Does sensor-level graph structure suffice?
Static graph	✓	matched	fixed edges	Does relational routing help without edge inference?
Inferred dynamic graph	✓	matched	observed/inferred	Does topology inference add value?
Oracle graph	✓	matched	privileged edges	What is the attainable structural upper bound?
Random / shuffled graph	✓	matched	invalid control	Does edge semantics, rather than regularization, matter?

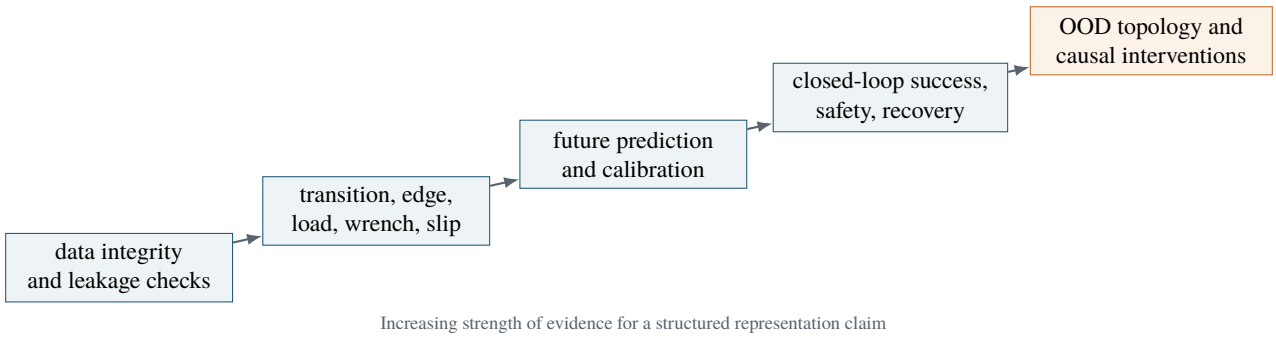


Figure 4: An evidence ladder for representation research. Higher rungs depend on, rather than replace, the lower-level checks.

For the tool-use phase, contact-centric alternatives are also mandatory: compare system-level interface edges with a continuous contact-field condition and a contact-anchored policy frame [17, 52]. These are phase-specific baselines rather than reasons to give the graph model extra geometry or labels.

A useful diagnostic is the fraction of oracle improvement retained by inferred structure. Let M be a higher-is-better metric and M_U, M_I, and M_O the unstructured, inferred, and oracle scores. Then

$$R_{\text{oracle}} = \frac{M_I - M_U}{M_O - M_U + \epsilon}. \tag{3}$$

This ratio is descriptive, not a universal target. It is unstable when the oracle gain is small and should always be reported with raw scores and uncertainty. For error metrics, the signs must be adjusted. Its value is that it distinguishes a weak edge estimator from an unhelpful structural hypothesis.

Match observation refresh and the number of open-loop executed actions as well as nominal action playback rate. SimpleTouch's matched-horizon ablations and VISTA's distinction between 10 Hz playback and a 0.6–0.8 s observation refresh show why fast action commands do not imply fast tactile feedback [88, 97]. A representation comparison that also changes feedback frequency cannot assign the full gain to its encoding.

8.3 Metrics aligned to the claimed mechanism

Contact and phase. Use per-class precision, recall, and macro-F1 for giver-only, dual-contact, receiver-only, no-contact, tool-contact, and slip states. Frame accuracy should be supplemented with transition timing error, segment-level F1, and early-warning horizon. Long stable segments otherwise dominate the score.

Edges. Dynamic-edge evaluation requires precision–recall AUC under class imbalance, onset/offset timing, false persistent edges, and calibration. Node/edge identity must be permutation-consistent where appropriate. Testing only edges used to construct training labels risks circularity; held-out contact configurations and sensor ablations are stronger.

Load and wrench. Report load-share MAE, absolute force/wrench MAE in a stated frame, angular error for force direction, normalized error across object masses, and horizon-specific future prediction. Estimated tactile wrench should not be described as independent ground truth. Gravity compensation, filtering, and coordinate transformations belong in the method, not supplementary guesswork.

Tactile in-hand rotation offers a concrete joint progress–load diagnostic [25]. A symmetric layout improves rotation while lowering pooled estimated load and load-sharing variation. However, its load sums tactile-estimated force magnitudes, not independently measured net wrench; hardware lacks tactile-input ablations. Simulation normalized rotation and hardware revolutions also use different units. These distinctions should accompany any transfer of the metric to bimanual release or stabilization.

Slip and uncertainty. Slip is an event-detection problem with severe imbalance. Precision–recall, false alarms per minute, and warning time are more informative than accuracy. Fine-grained contact-state and slip-direction sensing provides a concrete local classification reference [54], but local class accuracy should not be reported as evidence of cross-interface load inference. For uncertainty, use reliability diagrams, expected calibration error, Brier score, negative log likelihood, and risk–coverage curves. Calibration should be re-evaluated under object, sensor, and topology shift.

RATE adds a temporal-risk auxiliary-target comparator: an LSTM tactile history and training-only future-change/alert targets feed a residual representation adapter [35]. Its real tests use two sensors on one parallel-jaw gripper. The reported contact-safe success rate is safe successes divided by successes, with a deformation-depth threshold; it is neither an all-trial safe- success probability nor a force-based safety guarantee. Exact task-wise alert construction was not located in the

audited text. Future studies should publish that provenance and separately report success, unsafe outcomes, and calibration; a risk-related training objective alone does not establish calibrated uncertainty.

Duration, relaxation, and damage. Sustained-contact evaluation should stratify error by contact duration, temperature, material, and load/unload history; relaxation-aware force sensing shows why a single aggregate MAE can hide systematic drift [82]. For deformable objects, report task success and a separate deformation or damage constraint. SoftVTBench's deformation-aware success rate is one example of requiring both completion and bounded peak deformation [37]; the exact threshold and evaluator-only state must remain explicit.

Closed-loop outcomes. Success should be paired with time, peak force, impulse, drop rate, object damage proxy, path error, number of recovery events, and unnecessary grip. A structured model that achieves the same success with lower peak force or earlier safe release can still be valuable. Failure-aware execution further requires separate accounting for detection, local retry, reset/recovery, and irrecoverable failure. FLARE operationalizes these branches with an online monitor and distinct retry/reset mechanisms [114]; it is a visual–language recovery framework rather than a tactile representation, but it makes recovery-aware metrics a necessary control for long-horizon claims.

8.4 Splits and out-of-distribution tests

Random frame splits are invalid for temporally correlated episodes. The split hierarchy should include, where relevant:

- episode and contiguous contact-sequence holds-out;
- object identity, geometry, mass, friction, and compliance holds-out;
- grasp pose and hand-role swap;
- collection session, operator/demonstrator, and calibration session;
- sensor instance, sensor type, missing channels, and added latency;
- embodiment or morphology;
- task/contact topology, such as handover training and tool-use testing.

The split must be made before window extraction so adjacent windows cannot cross train/test boundaries. Hyperparameter selection should use a validation split with the same unit as the intended generalization claim.

The 30-paper full-text core contained zero studies that met a strict contact-topology OOD criterion under deployment-observable inference. Strict topology OOD holds out the contact-participant or edge structure itself (for example, hand–object–hand training followed by hand–tool–workpiece–hand testing). Unseen object instances, new tool geometry, contact-mode variation within a fixed participant structure, and semantic tool–action composition are valuable tests, but they must be reported separately rather than relabelled as topology transfer.

8.5 Causal and counterfactual interventions

Correlational probes show what can be decoded; interventions show whether the claimed structure matters. Low-cost interventions include time-shuffling one hand's tactile stream, swapping left/right roles, masking one fingertip, delaying receiver contact, perturbing the workpiece, rewiring predicted edges, or freezing edge probabilities. Physical interventions include known mass/friction changes, brief stabilizer motion, tool stiffness changes, or an unexpected contact. The expected direction of change should be stated before the experiment.

For temporal baselines, mask history content, shuffle its order, and disable observation-driven latent correction exactly rather than replacing the model with an unrelated architecture. TemporalFlow-VLA and PredVLA make these interventions especially concrete [72, 101]. For hierarchical tactile prediction, compare flat heads, stop-gradient removal, and dependency-order shuffling [96].

For example, delaying receiver contact should delay load-share crossover and safe release. If a model's phase prediction is unchanged, it may be following a motion schedule rather than tactile evidence. Rewiring a tool–workpiece edge should damage future-wrench prediction more than rewiring an inactive relation. If all rewirings have equal effect, the graph may act as generic regularization.

CALIPER adds a more basic validity check: an intervention must change the physical quantity the representation is supposed to identify [58]. Equal readout budgets, random-feature floors, calibration-history swaps, and random-direction erasure controls expose clean-scene shortcuts that disappear under nuisance variation. Its evidence is simulated fixed-contact-type inference and one-step speed selection, not closed-loop or topology-transfer performance. For tactile studies, analogous tests should verify perturbation effect sizes, compare raw/random encoders, and swap plausible force histories before interpreting a decoded physical variable as a mechanism used by the controller.

Physical consistency also requires trustworthy sensing. PhantomTac shows that controlled interference can corrupt magnetic tactile force estimates and disrupt feedback [110]. This motivates independent cross-modal checks and sensor-specific robustness tests before interpreting an inferred edge as physical contact. The proposed proprioceptive consistency defense is not quantitatively validated, and the findings should not be generalized to optical, capacitive, or resistive touch without corresponding experiments.

8.6 Statistics and reproducibility

Formal results should use at least five independently seeded runs where training variance is material; early internal decisions can use three seeds but must not be reported as definitive. Pair results by the same data split and environment seed. Report effect sizes and confidence intervals, not only significance tests. For episode success, use bootstrap or binomial intervals with episode as the unit; for dense time series, aggregate within episode before testing or use hierarchical resampling.

Reproducibility requires a data lineage from raw sensor packet to metric: hashes or version identifiers, synchronization configuration, calibration, split manifests, normalization fitted only on training data, model configuration, seeds, software environment, checkpoint selection, and evaluation script. Code audits should distinguish a paper's claims from what a public snapshot executes, as illustrated by the PhysGraph case in section 6.

Table 8: Claim-to-evidence checklist. A paper need not include every row, but the evidence should match its stated contribution.

Claim	Minimum comparison	Stronger evidence
Touch is necessary	proprio/vision-only and tactile ablation	sensor masking and latency interventions
Explicit structure helps	matched attention/temporal/static graph	random-edge control and oracle–inferred analysis
Edges are observation-grounded	no test-time privileged inputs	pose/sensor corruption and edge calibration
Representation is physical	interface/load predictive targets	force-balance residuals and counterfactual response
Generalizes	object/session/contact-sequence split	sensor, embodiment, role, and topology holds-out
Improves control	multi-seed success and safety metrics	disturbance recovery and risk–coverage analysis

9 Synthesis and Research Agenda

9.1 What the literature currently supports

The mapped evidence supports six qualified conclusions.

First, tactile-specific pretraining is useful. Across robotic play, self-supervision, multimodal alignment, and cross-sensor learning, tactile representations outperform purely task-specific training in several downstream settings [20, 23, 28, 100]. This justifies reusing a strong local encoder rather than learning raw tactile features from scratch for every bimanual experiment.

Second, spatial and relational inductive biases can be useful when the task has known structure. Morphology graphs, tactile graphs, object-centric dynamics, and contact fields have succeeded in in-hand manipulation, physics prediction, packing, and contact tracking [1, 21, 27, 102]. This establishes plausibility, not universal superiority over attention.

Third, bimanual dexterity has become an active, increasingly crowded research area. Large simulation suites, demonstration-generation systems, retargeting, and real policies now cover many complex tasks [9, 36, 43, 74]. Novelty must therefore be located in a specific mechanism or evaluation gap.

Fourth, contact-centric and tactile-reactive control is moving toward broader embodiments, tools, and data scales [52, 55, 63, 107, 119]. Because many examples are recent preprints or programme-listed works, they define a competitive frontier more than a settled empirical consensus.

Fifth, the strongest competing explanations are increasingly structured even when they are not tactile graphs. Directed tactile prediction [96], mesh-prior plus diffusion-residual dynamics [84], temporal-flow supervision [101], prediction-error recurrence [72], and failure-aware recovery [114] each isolate a different source of performance. This makes matched objective, history, and recovery controls part of the minimum evidence for an explicit contact-structure claim.

Sixth, direct evidence now reaches beyond isolated local tactile features. BiView-Touch supports useful cross-hand predictive dependence in human recordings; causal force-history plus compliance supports real bimanual release/hold

decisions; and CADeT supports visually inferred transmission modes with active probing [30, 46, 56]. These results narrow the gap. They do not jointly demonstrate independent robot load-path validity, matched structural benefit, calibrated uncertainty, and held-out contact-participant topology. Meanwhile, ContactWorld and the October SimpleTouch/VISTA controls make representation, goal information, and feedback frequency separate competing explanations [88, 97, 113].

9.2 What the literature does not yet establish

The evidence does not yet establish that an inferred dynamic contact graph is better than a matched temporal Transformer for bimanual load transfer. It does not establish that local tactile foundation models preserve cross-interface wrench information. It does not yet establish robust, uncertainty-tested recovery of robotic load-path topology from deployable touch and proprioception across changed participant structures. It also does not provide a standard benchmark for transfer from handover to cooperative tool use.

These gaps are linked by an evidence chain:

$$\begin{aligned}\text{sensor signal} &\rightarrow \text{local contact factor} \rightarrow \text{entity/interface identity} \\ &\rightarrow \text{dynamic relation} \rightarrow \text{future physical outcome} \rightarrow \text{control decision}.\end{aligned} \tag{4}$$

Current papers often validate one or two adjacent links. A local encoder validates the first link; a privileged graph policy validates the last links conditional on correct structure; a successful bimanual imitation policy validates behaviour but may leave the intermediate mechanism unidentified. A high-impact study need not solve the complete chain, but it should state exactly which link it tests.

9.3 Eight priority problems

Table 9: Prioritized research agenda for structured bimanual tactile representations.

P	Problem	Decisive experiment	Potential deliverable
1	Observation-grounded dynamic interfaces	infer hand–object edges from tactile, proprioception, and optional vision; compare with oracle and random edges	contact-inference benchmark and calibrated edge model
2	Cross-interface physical targets	predict phase, load share, future wrench, slip, and relaxation state across short horizons and sustained contact	probe suite and physically annotated handover data
3	Matched structural comparison	compare concat, attention, directed/flat prediction, prediction-error recurrence, graph-only, residual-only, inferred, and oracle structure with equal budgets	defensible evidence for or against explicit structure
4	Cross-topology transfer	handover pretraining followed by tool–workpiece adaptation under label/data budgets	compositional transfer method and benchmark
5	Uncertainty under sensor failure	missing taxels, saturation, wear, thermal/viscoelastic drift, latency, and sensor-type shift	risk-aware contact representation
6	Privileged-to-observation distillation	train with simulator contacts/wrenches; deploy an observation-only student	scalable supervision without deployment leakage
7	Causal structure tests	role swap, receiver delay, history-order/edge rewiring, physical perturbation, and retry/reset separation	evidence that relations mediate predictions/control and recovery
8	Interoperable data lineage	common schema, split manifests, timestamps, frames, masks, release versions, and label provenance	reproducible BiLoad-style evidence matrix and data tooling

9.4 A staged experimental programme

The literature suggests a risk-reducing order of work.

Stage 1: audit and offline probing. If a downloadable, licensed T-Rex subset is verified, audit 20–100 episodes for timestamps, missing channels, task phases, object identity, and label provenance [63]. Otherwise, make the VTDexManip bimanual handover task the first executable benchmark while T-Rex ingestion remains a parallel artifact audit [49]. Reproduce PhysGraph as stock public code and a separately named corrected-bias variant; do not merge the two interpretations [44]. The output should be a unified schema and an initial proprio-only/concat/temporal probing table, not a large control policy.

Stage 2: representation isolation. Fix one local tactile encoder and compare matched fusion/structure variants on phase, future edge, load share, future interface wrench, slip risk, and uncertainty. The matrix should include a flat versus directed tactile hierarchy, prediction-error recurrence, training-only temporal grounding, graph-only versus global-residual-only models, and their structured combination [72, 84, 96, 101]. Match observation refresh and executed action horizon as well as representation budgets. Include independent load labels and a dense cross-hand fusion control; neither phase proxies nor attention maps establish force transmission. Use episode-, object-, and contact-sequence splits, plus contact-duration and temperature strata where sustained loading is present. Early project decisions may use three seeds, but final claims should use five or more where variance warrants. Internal go/no-go thresholds may guide resource allocation, yet they must not be written as achieved results or retrospectively optimized paper hypotheses.

Stage 3: closed-loop handover. Only after the offline model beats strong temporal baselines should it enter closed-loop quasi-static handover. Compare fixed schedules, causal force-history with compliance, unstructured tactile feedback, anchored tokens, cross-hand completion, and inferred structure [46, 56]. Evaluate safe release, peak force, drop rate, time, uncertainty, failure detection, local retry, and reset under receiver delay and object-property changes. Recovery branches should be reported separately so that controller recovery does not masquerade as a representation improvement [114].

Stage 4: gated cooperative tool use. Freeze a rigid stylus or wiping-puck–plate task early, but move to full closed-loop experiments only if (i) representation gains survive matched budgets and OOD splits, and (ii) the coordination-necessity check confirms that the stabilizing hand materially affects the outcome. The second study should emphasize transfer from a three-node handover path to a longer tool–workpiece path. If the gate fails, a valuable pivot is privileged-to-observation graph distillation or an interface-inference benchmark rather than an underdiagnosed hardware demo.

9.5 Where a bimanual contact-graph encoder can contribute

An encoder built around bimanual contact flow remains scientifically meaningful, but only under a narrow and testable formulation. Its strong contribution would not be "a GNN for two tactile hands." It would be the combination of:

- hierarchical sensor/link/entity nodes that preserve quality masks and coordinate frames;
- dynamic, probabilistic edges inferred from deployable observation;
- object- or task-frame messages that represent load propagation through objects and tools rather than a permanent left–right shortcut;
- self-supervised or privileged-distilled edge/physics targets;
- calibrated uncertainty and explicit missing-sensor behaviour; and
- matched-budget evidence that the factorization improves transitions, data efficiency, or cross-topology generalization.

Under this formulation, the encoder is a major hypothesis and potential contribution. Without observation-grounded edges and decisive comparisons, it is an implementation variant in a crowded graph-and-touch landscape.

9.6 Publication opportunities

The research agenda naturally separates into two publishable units. The first is a benchmark-and-representation paper: synchronized handover data and probes, matched structural baselines, dynamic interface inference, and careful observability audits. The second is a transfer paper: reuse of interface/relation functions from handover in cooperative tool use, with role swap, new nodes, limited-data adaptation, and causal perturbations. A third, lower-hardware-risk alternative is an artifact paper on privileged-to-observation contact inference and reproducible code auditing across simulators.

The literature review itself could support publication after a more systematic update: preregistered database searches, deduplication across indexes, dual screening, complete author/publication metadata, code/data-availability coding, and a

public evidence table. The current manuscript supplies the conceptual taxonomy and reproducible local scaffold, but it should not be labelled a systematic review until those steps are completed.

10 Limitations and Conclusion

10.1 Limitations of this review

This review is intentionally broad but not exhaustive. The archive was assembled for a concrete research decision, which biases it toward dexterous hands, learning-based control, recent tactile foundation models, and graph/contact representations. It under-samples classical grasp mechanics, human haptics, teleoperation, non-learning force control, and some industrial dual-arm systems. The synthesis does not use dual independent screening or extraction. The 86-PDF cache is an August historical record; the 7 October pass verified three new PDFs; the 14 October pass uses fourteen official HTML full texts and two official PDFs, without a full cache refresh. The 21 October pass adds fourteen HTML audits and one PDF audit, with two existing-paper version comparisons. Five manifest entries remain source-only, and publication/code/data status can change rapidly. Several 2026 methods are preprints or accepted/programme-listed works; their final claims and artifacts may differ. Thirty purposively selected core papers received full-text precision coding, while 105 scholarly records remain outside that schema: 45 have located full-text archive audits across the October updates and 60 retain the older lighter archive-wide coding layer. The precision subset is not a random sample, and neither layer used dual independent extraction or inter-rater agreement. Transparent evidence cards reduce ambiguity but do not replace those controls or independent reproduction.

The corpus is too heterogeneous for meta-analysis. Reported success rates mix different embodiments, sensors, training budgets, objects, horizons, and definitions. Counts therefore describe only the curated archive. The review also does not claim that load paths are the only useful abstraction. Deformation, compliance, friction cones, contact fields, hybrid modes, and action-conditioned latent dynamics may be superior in particular regimes.

The archive was initially frozen at 74 records on 12 August, updated to 91 on 30 August, and refrozen at 105 on 7 October 2026. That catch-up added 13 in-window first releases and one older work discovered through a September revision. Metadata-only watchlist leads and an unread VT-MUSE revision remain outside the new synthesis. This improves recency without making the search exhaustive. No surveillance addition was retroactively treated as part of the 30-paper precision subset. The 14 October surveillance integrates 16 additional full-text archive records and refreezes the narrative at 121; the precision denominator and its historical date remain unchanged. The 21 October pass adds 15 full-text audits and two existing-paper revision checks, integrates their interpretation, and refreezes the manuscript at 136 records.

10.2 Conclusion

Tactile robotics and bimanual manipulation have each advanced rapidly, but their intersection exposes an incompletely validated representational layer. Local tactile encoders identify what happens at a sensor; bimanual coordination requires a model of how several interfaces become coupled through shared entities and how that coupling changes at contact transitions. Graphs, spatial anchors, contact fields, modes, and unstructured temporal attention are competing ways to express this layer. None should be presumed best.

The August additions made the comparison more demanding. Directed tactile prediction, sequential visuo-tactile fusion, physical-property latents, mesh-prior residual dynamics, visual world models, prediction-error recurrence, and failure-aware recovery now provide stronger alternative explanations for performance. A contact-flow model must isolate its benefit from those mechanisms and show that the benefit comes from deployable cross-interface evidence.

The new full-text evidence requires a substantive narrowing. Cross-hand tactile representation learning is already demonstrated in offline human recordings; causal force history plus compliance is an effective real bimanual handover baseline; and visual transmission-mode inference with physical probing is already possible [30, 46, 56]. The unresolved contribution is the joint validation of observation-grounded robotic cross-interface loads, matched representation benefit, and transfer under a declared change in contact participants or edges. Neither an offline predictive dependency nor a predefined mode posterior alone establishes that conjunction.

The world-model and early-October evidence also weakens architecture-only claims. Predictive tactile supervision can help even when its output is absent from control, and strong scores can coexist with little measured dependence on current touch [45, 97, 111]. Geometry-preserving world-model controls and pose-conditioned equivariant fusion must be matched for sensor information, goals, and feedback frequency [88, 113]. A generated tactile future, a good surrogate evaluator, or a risk-related objective is therefore a hypothesis to test, not direct evidence of physical load propagation or calibrated safety [35, 68].

The 14 October evidence reinforces, rather than reverses, this narrowed claim. Temporal-role conditioning, predictive guidance, latent action targets, contact references, stage-specific force experts, and directional compliance supply stronger alternatives [11, 12, 34, 61, 108, 117]. Real cooperative tool execution is present, but does not by itself validate independently measured cross-interface loads. Canonical human-touch alignment and larger datasets must also be separated

from matched-budget online tactile dependence. Future evaluations should match reference access, auxiliary experience, routing assistance, feedback timing and controller compliance before attributing a gain to contact structure.

The 21 October evidence further strengthens compact tactile-feedback, local bimanual-control, and multimodal world-model baselines. Same-checkpoint tactile removal in blind Baoding supports runtime dependence on a small binary signal; ContactPolicy and PhysCanvas establish additional real bimanual evidence, not independently calibrated load-path inference [3, 18, 91]. DexGraspFlow's unseen finger-removal tests are important kinematic-topology transfer, distinct from a declared holdout of inferred contact participants or edges [118]. These findings retain the narrowed joint gap, but require a more explicit separation of morphology, sensed contact, control cadence and pretraining budget. ForeTac v2 changes its quality objective rather than its headline outcomes; ILA v2 removes a premature conference header, so neither revision justifies a publication-status upgrade.

The strongest immediate benchmark is quasi-static handover because it provides observable phases and controlled load redistribution. Cooperative tool–workpiece manipulation is an established but complementary extension that adds a longer and role-asymmetric load path. The narrower question of matched, observation-grounded cross-hand load flow is comparatively sparse in the mapped corpus; it belongs after the simpler mechanism has passed matched-budget and coordination-necessity gates. Across both tasks, the decisive methodological requirements are deployment-observable inputs, explicit label provenance, episode/object/contact-sequence splits, oracle–inferred–random structure comparisons, uncertainty, and interventions.

The full-text core reinforces this requirement: zero of 30 papers evaluated strict held-out contact-topology transfer under deployment-observable inference. This remains a historical scoped result for that fixed subset, not a zero count for the 135 scholarly works in the expanded archive or the entire literature, and it motivates a decisive experiment rather than a priority claim.

An explicit bimanual contact-flow encoder can therefore be a strong contribution, but the graph layer is not the contribution. The contribution is a falsifiable factorization of sensed interfaces and object-mediated load propagation, together with evidence that this factorization improves prediction, control, or transfer under conditions where simpler temporal fusion does not. That framing provides a clear starting point for experiments and a durable standard against which future structured tactile representations can be judged.

A Review Update Record

This appendix records changes to the review corpus and interpretation after the initial manuscript freeze. It prevents three dates from being conflated: the archive snapshot used by the narrative, the completion date of the 30-paper precision audit, and the most recent surveillance date. Detailed screening notes remain in the dated files under `weekly_updates/`; this appendix records only changes that were integrated into the main manuscript.

Table 10: Version history of the review corpus and synthesis.

Date	Corpus state	Review action	Effect on interpretation
12 Aug. 2026	74 records; 69 local PDFs; five source-only	Initial narrative freeze after scope, status, and archive-wide coding	Established the six-axis taxonomy and the observation-grounded cross-interface gap.
13 Aug. 2026	30-paper purposive core: 19 Level A, 11 Level B	Completed full-text precision coding and evidence cards	Established the scoped result that 0/30 tested strict held-out contact topology under deployment-observable inference.
26 Aug. 2026	Added 10 records; archive 84; 79 local PDFs	Rapid surveillance for 20–26 Aug.; archive-wide coding and artifact audit	Added directed tactile prediction, stronger sequential fusion, physical-property probes, local slip sensing, contact-event controls, bimanual robustness/geometric baselines, and deformation-aware safety.
30 Aug. 2026	Added seven records; archive 91; 86 local PDFs	Partial-week surveillance for 27–30 Aug.; archive-wide coding and artifact audit	Added structured-prior plus global-residual dynamics, relaxation-aware force sensing, cross-embodiment/online bimanual dynamics, temporal-flow and prediction-error baselines, and explicit retry/reset recovery.

Date	Corpus state	Review action	Effect on interpretation
30 Aug. 2026	91 records: 90 scholarly, one resource; precision core unchanged at 30	Integrated both surveillance rounds into the main narrative and refroze the manuscript corpus	Strengthened the baseline and evaluation requirements; did not change the central gap judgment or the precision denominator.
7 Oct. 2026	Added 14 records; archive 105: 104 scholarly, one resource; precision remains 30	Full-text archive audit and narrative integration of 13 new releases plus one older-work revision discovery; three selected PDFs separately verified	Narrowed the gap in light of cross-hand completion, temporal-compliant handover, and observed transmission modes; strengthened feedback, goal, metric, and intervention controls.
14 Oct. 2026	Added 16 records; archive 121: 120 scholarly, one resource; precision remains 30	Located full-text audit and narrative integration: 15 first releases plus TouchScale revision discovery; fourteen HTML texts and two PDFs	Strengthened temporal, predictive, stage-force, compliance and data-scale controls; retained the narrowed joint gap with explicit timing, routing, metric and artifact boundaries.
21 Oct. 2026	Added 15 records; archive 136: 135 scholarly, one resource; precision remains 30	Full-text narrative integration plus ForeTac/ILA revision comparison; boundary leads kept separate	Retained the narrowed joint gap; strengthened binary-feedback, local bimanual, world-model and kinematic-topology controls; corrected ForeTac scorer without claiming new outcomes.

A.1 Studies integrated from the 20–26 August update

The first surveillance round added ten studies. HiTac-WAM and VT-MUSE directly strengthened predictive and sequential visuo-tactile baselines [92, 96]. ViTacPhys, fine-grained slip sensing, and a durable tactile fingertip expanded physical-property, local-event, and hardware- reliability considerations [16, 50, 54]. Explicit physical-interaction events and Co-ToGrasp clarified the distinction between demonstrated or prescribed contact structure and topology inferred from deployment observations [59, 66]. Modality masking and PartialBiGrasp strengthened non-tactile bimanual controls [13, 39], while SoftVTBench added a safety-aware deformable-object benchmark [37].

A.2 Studies integrated from the 27–30 August update

The second round added seven studies. MeshPriorDiT motivated a three-way comparison among structured prior, expressive residual, and their hierarchy [84]; relaxation-aware soft-gripper sensing motivated contact-duration and thermal/viscoelastic drift tests [82]. Robot juggling and CLAP strengthened rapid online-dynamics and cross-embodiment video-world-model controls without supplying distributed touch [41, 48]. TemporalFlow-VLA and PredVLA motivated history-order interventions and exact open-loop prediction-error ablations [72, 101]. FLARE made failure detection, local retry, and reset/recovery separate evaluation outcomes [114].

A.3 Interpretive and artifact boundaries retained

The integrated update changes what constitutes a credible baseline, but it does not upgrade every record to full-text precision evidence. In particular:

- "contact topology" in grasp synthesis can denote a requested static layout; it is not automatically a dynamic, observation-inferred task graph;
- geometry-supervised temporal history, visual world models, and visual– language recovery remain non-tactile controls unless distributed touch is an actual deployment input;
- the ViTacPhys artifact remained forthcoming at the audit date, while the public SoftVTBench release and current manuscript described different dataset scales; and
- CLAP provided a public code/checkpoint entry, but artifact availability does not by itself establish tactile or load-topology evidence.

Future updates should append a dated row, record changes in publication and artifact status, and state explicitly whether the manuscript corpus, the precision subset, or both were changed. Counts based on the 30-paper precision audit must retain that denominator until new papers receive the same schema, located-evidence card, and quality checks.

A.4 Studies integrated in the 7 October catch-up

This pass covers sources available on 7 October for first releases after 30 August, plus ContactWorld, first posted in June and discovered through its September revision. Fourteen new archive records received located full-text audits and narrative integration; the 30-paper precision schema and its August completion date are unchanged. The manuscript corpus is refrozen at 105 records. The 86-PDF August cache remains a historical snapshot: only CALIPER, CADeT, and DemoMimic PDFs were separately validated in this pass; other new full texts were read in official HTML. Metadata-only leads and the inaccessible current VT-MUSE revision were not used to change study-level claims.

BiView-Touch and temporal-compliant handover narrow the previous gap statement by providing direct, differently scoped cross-hand representation and control evidence [46, 56]. CADeT adds observable transmission-mode inference with active probing, while DemoMimic and Bench2Dex strengthen geometry and benchmark controls [30, 73, 104]. Dex-X makes retained reference conditioning explicit [7]. ME-Dex 1.0, DexTouch-WM, UVTA, and ContactWorld distinguish tactile supervision, future conditioning, representation information, and actual feedback dependence [45, 68, 111, 113]. RATE separates proxy-safe success from calibrated physical safety; CALIPER strengthens perturbation-validity controls [35, 58]. SimpleTouch and VISTA, both first posted on 2 October, add strong pretrained-token and pose-conditioned spatial controls while exposing feedback-frequency and finite-group assumptions [88, 97].

All 14 additions are coded as preprints or author-claim records. Workshop presentation, submission, and author-reported acceptance were not upgraded to independently verified venue status. The inherited TactiDex author-page acceptance is also reassigned to the author-claim tier, retaining its raw status. The conclusion now targets the conjunction of deployable robotic load evidence, matched structural benefit, and contact-topology-defined transfer, rather than absence of relational touch or inferred transmission modes. No expanded-corpus zero count is asserted.

A.5 Studies integrated in the 14 October surveillance

Sixteen new archive records receive located full-text audits. Fifteen first releases fall after 7 October; TouchScale first appeared on 7 October and is included as an 8 October revision discovery, without asserting a verified version-to-version scientific change. CoToGrasp's 8 October revision was checked without finding a substantive body/appendix change; it is not a new study. The narrative is refrozen at 121 records, including 120 scholarly works. The 30-paper precision subset remains dated 13 August.

TACROSS and TouchScale strengthen canonicalization and data-scale comparisons; PACE, DeTac, TacSpin, ForeTac and minimal tactile flow sharpen temporal, signal-specific, geometric and predictive controls [6, 11, 12, 42, 57, 90, 112]. CTP, ForceWAM and ILA distinguish reference contact, generated futures, latent-action supervision and actual online feedback [61, 95, 117]. Bimanual RMA, DEFT-VLA and FOMO-Dex add tracker-assisted adaptation, stage-specialized force control and directional compliance [34, 38, 108]. FEM-Tactile-GNN, the edge-native dual expert and MorphoPlanner expand local/body structure comparisons while retaining attribution and deployment limitations [76, 80, 116].

All new entries remain preprint/author-claim records. Fourteen were read in version-specific official HTML; ForeTac and MorphoPlanner used official PDFs. The full historical PDF cache was not refreshed and third-party PDFs are not shared in Git. The central joint gap is retained; stronger alternatives and more precise evaluation requirements are the substantive conclusion change. No expanded-corpus topology-zero count is claimed.

A.6 21 October integration and revision boundaries

Fifteen new studies receive located full-text archive audits, not precision-core promotion. Fourteen official HTML texts and one PDF support the additions; ForeTac and ILA exact-version PDFs support separate revision comparisons. The manuscript now uses 136 records, while precision coding remains dated 13 August with 30 papers. The dated ledger separately records same-day boundary and unaudited leads, selected version-history coverage, and artifact limits.

The central joint gap is retained, with stronger positive comparators: binary-feedback necessity, real local bimanual control, outcome-aligned tactile world modelling, and genuine kinematic link-removal transfer are now explicit. None is silently relabeled as matched, independently measured cross-interface load inference plus contact-topology transfer. Pre-training, online touch, demonstration context, future recorded signals and control cadence receive separate interpretations. ForeTac v2 changes four-class quality scoring to binary good/bad scoring without changing reported tables; ILA v2 only removes a premature conference header from otherwise matching extracted scientific text. Both remain preprints; the historical 14 October audit is preserved.

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